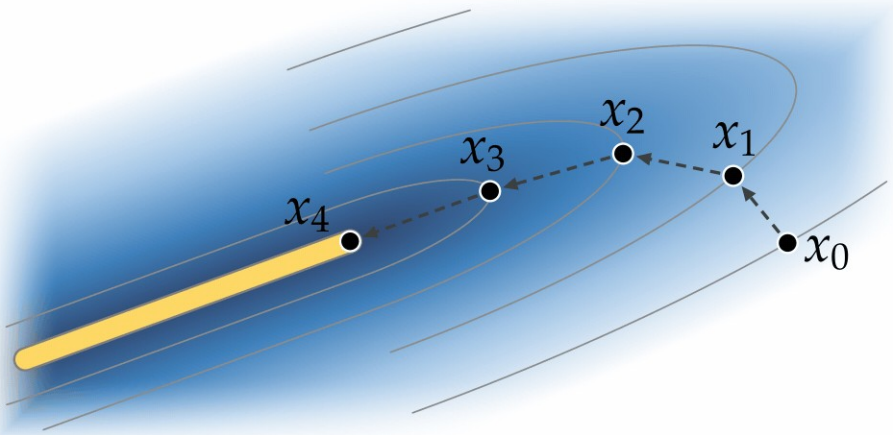
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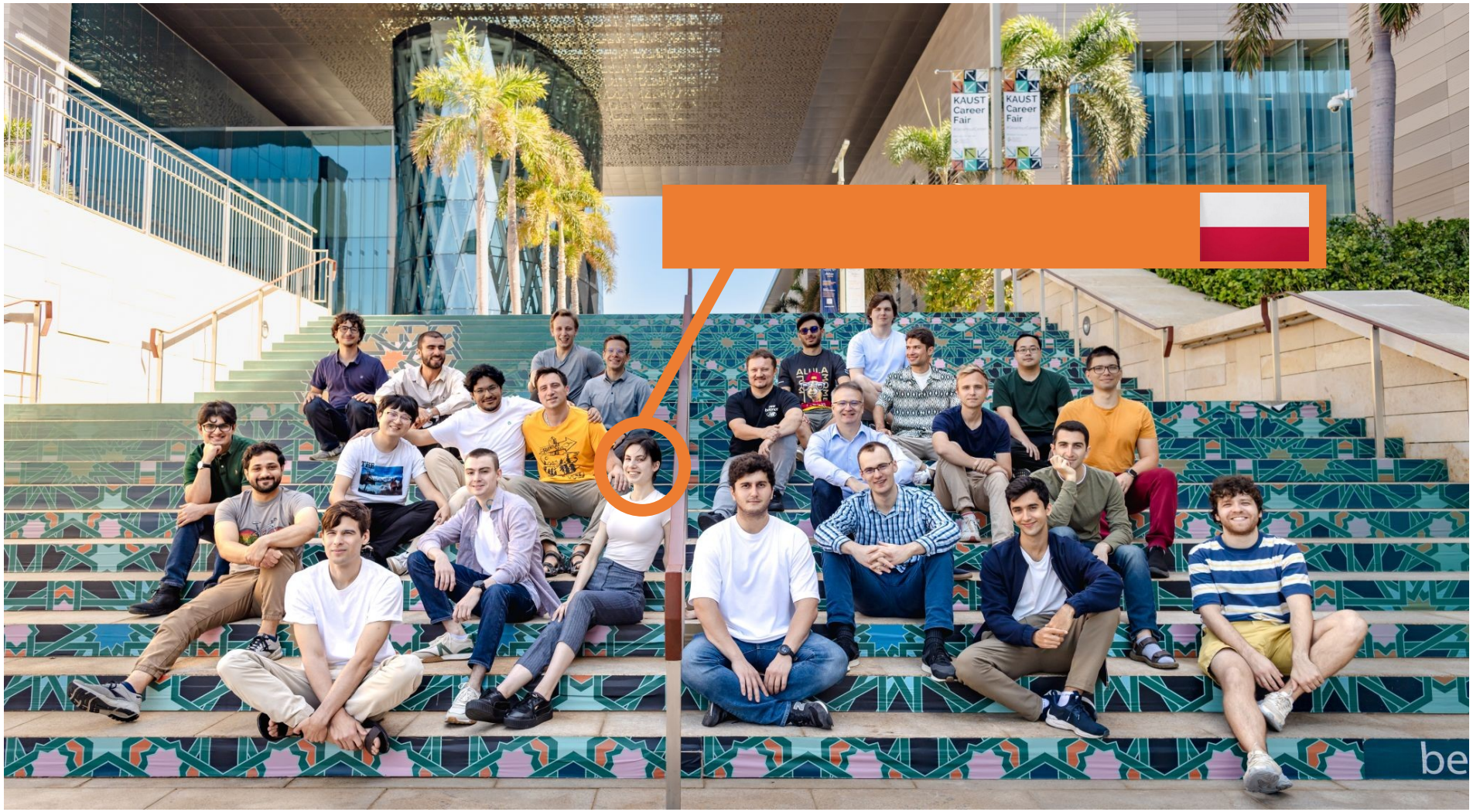
Guided by her advisor, Prof. Peter Richtarik, Kaja's innovative research bridges mathematics, statistics, and AI, advancing both the theory and practice of optimization for machine learning to make AI training faster, more cost-effective, and more resource-efficient.

She shares: "My goal is to bridge rigorous mathematical theory with practical solutions that scale to today's trillion-parameter AI models. This fellowship will help turn these ideas into tools that keep AI breakthroughs within reach."

At KAUST, we are proud to empower the next generation of AI leaders and drive innovation from Saudi Arabia to the world.



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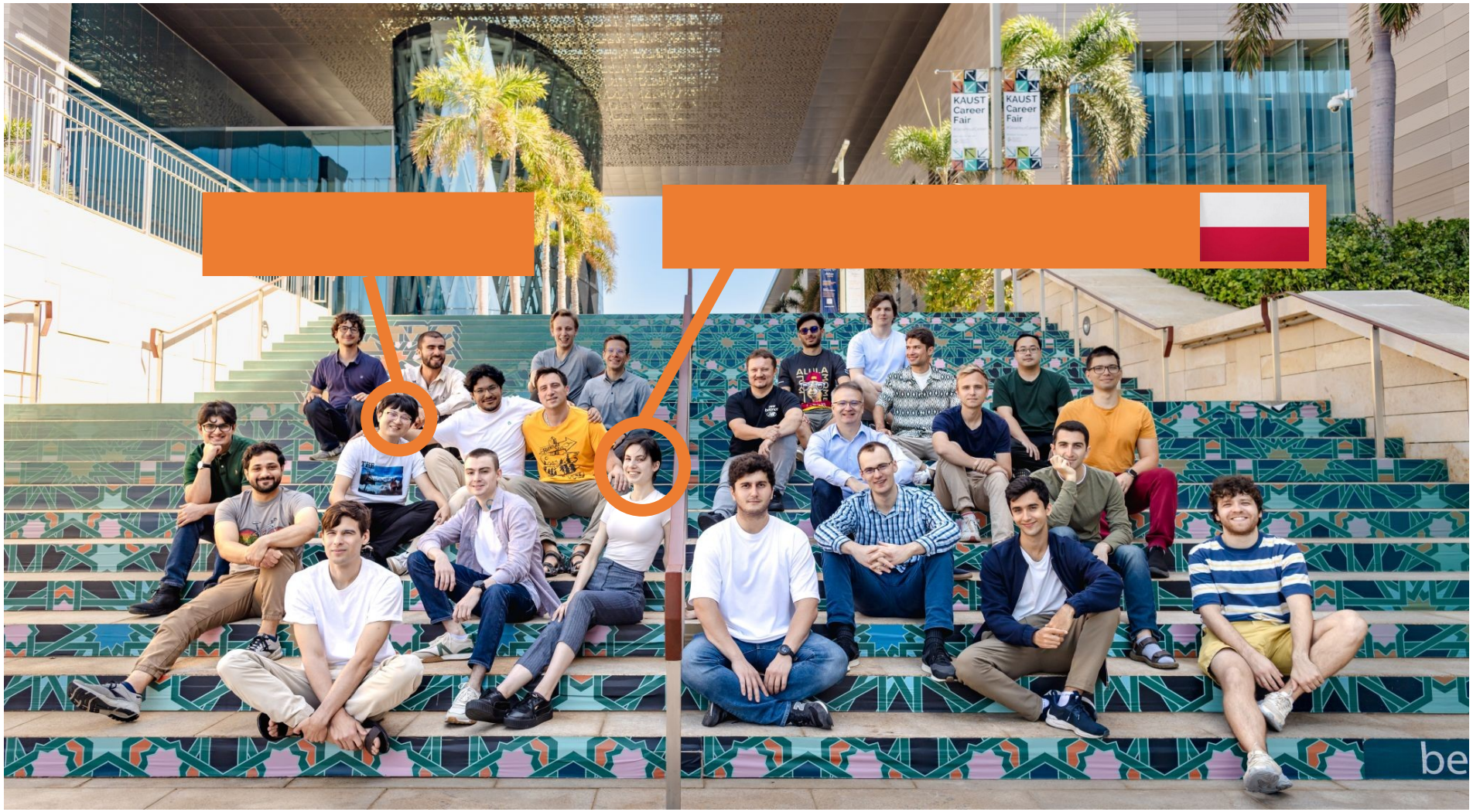
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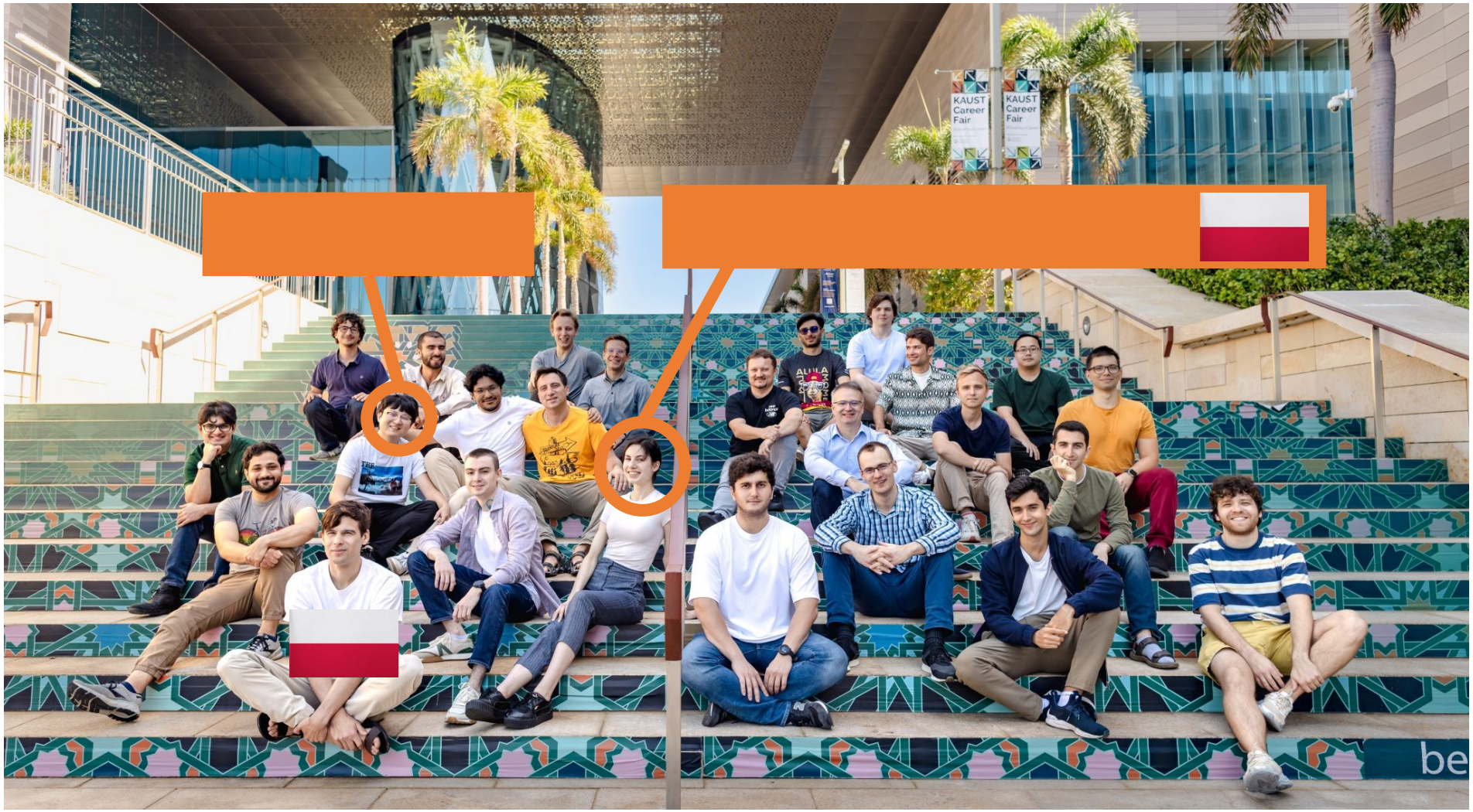
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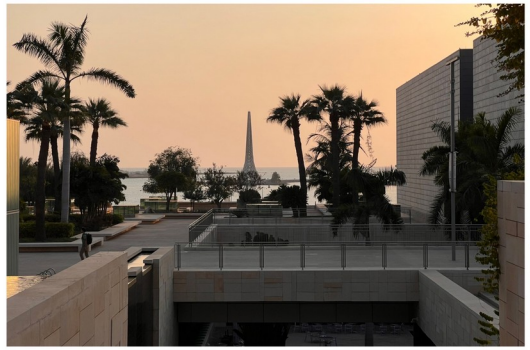
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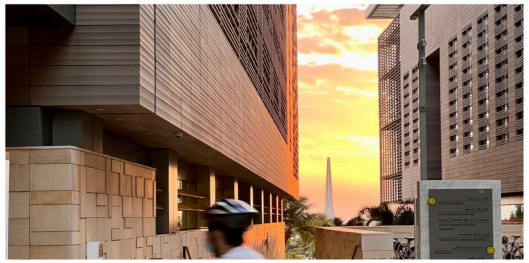


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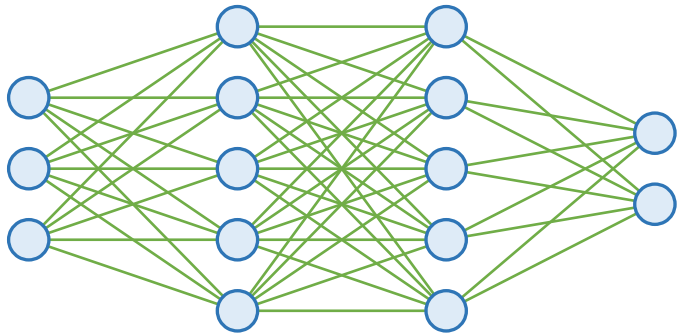






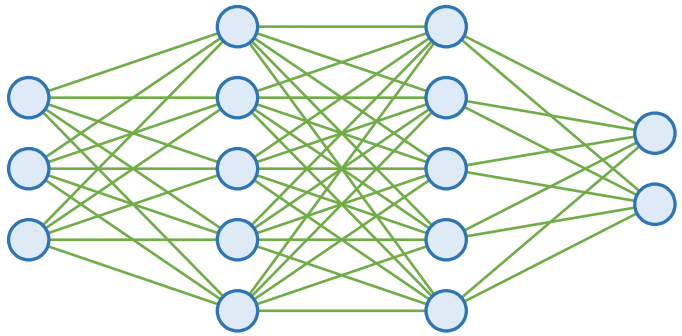


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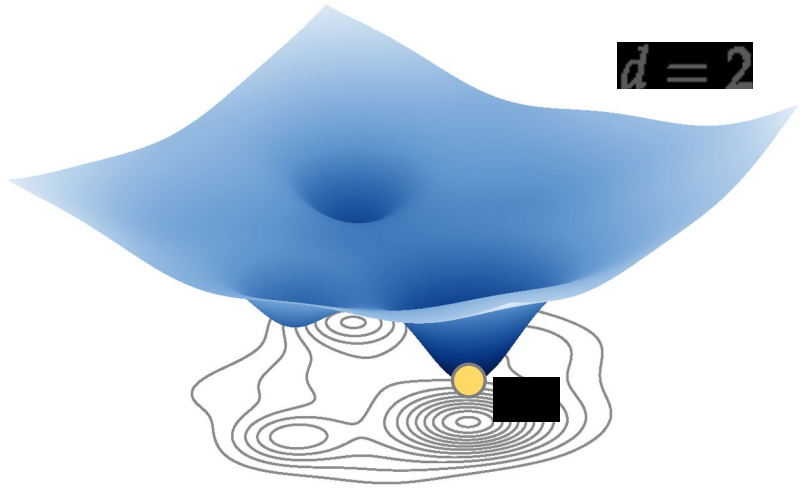
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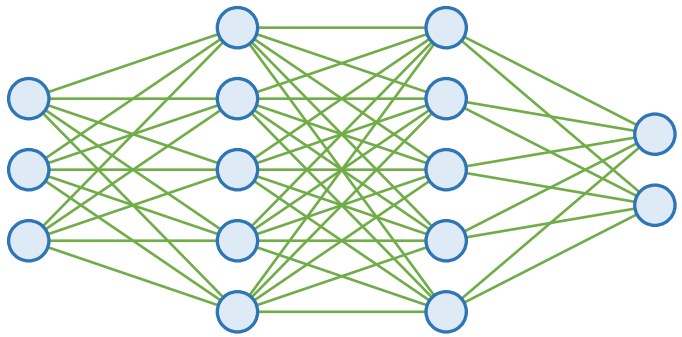
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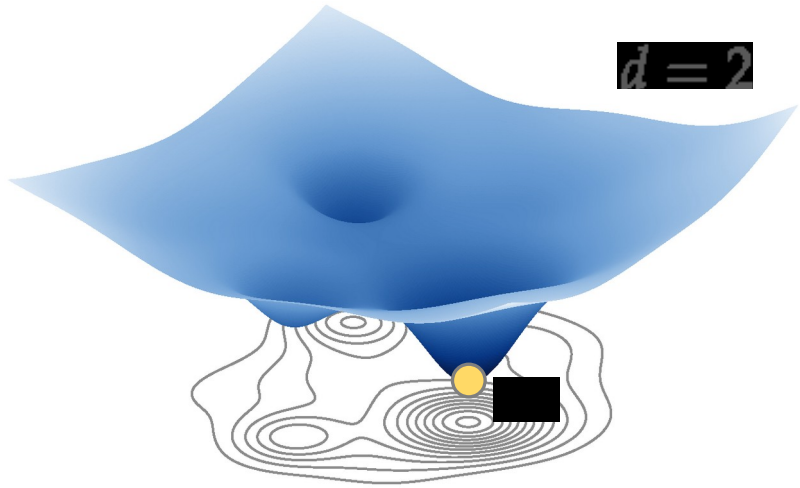
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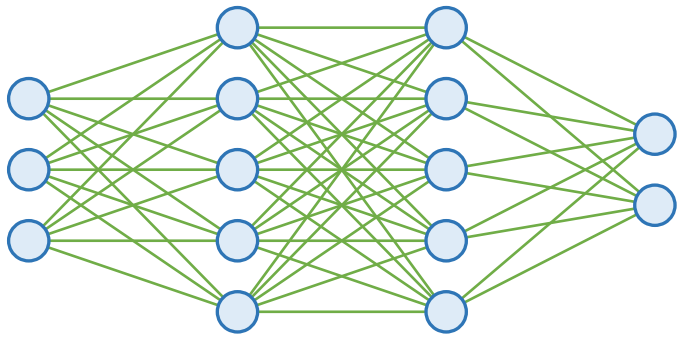




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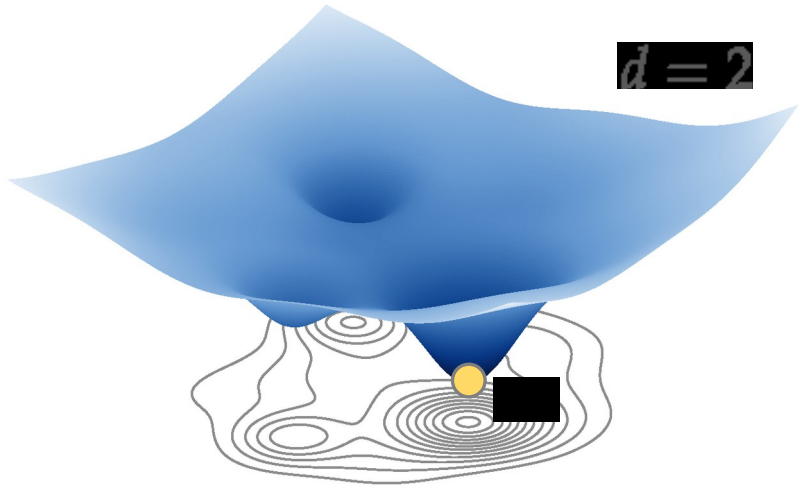
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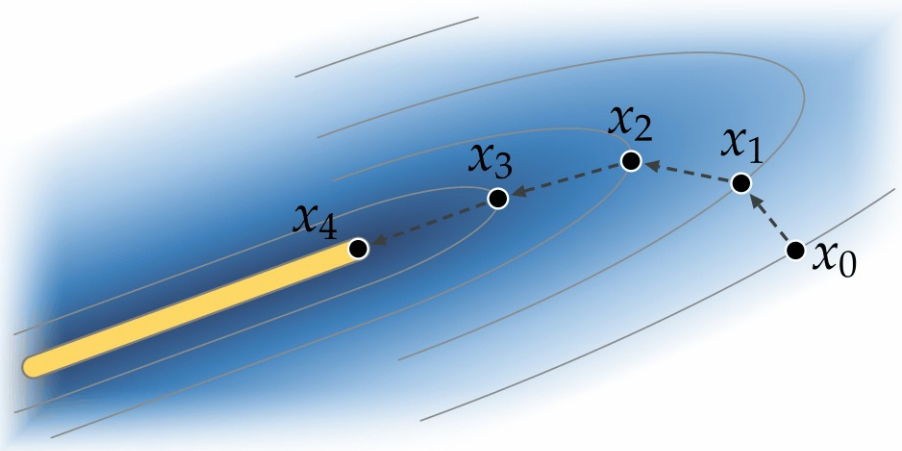
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arXiv:2003.08078v1 [math.OC] 18 Mar 2020

Acceleration with a Ball Optimization Oracle

Yair Carmon* Arun Jambulapati* Qijia Jiang* Yujia Jin* Yin Tat Lee†
Aaron Sidford* Kevin Tian*

Abstract

Consider an oracle which takes a point x and returns the minimizer of a convex function f in an ℓ_2 ball of radius r around x. It is straightforward to show that roughly $r^{-1}\log\frac{1}{\epsilon}$ calls to the oracle suffice to find an ϵ-approximate minimizer of f in an ℓ_2 unit ball. Perhaps surprisingly, this is not optimal: we design an accelerated algorithm which attains an ϵ-approximate minimizer with roughly $r^{-2/3}\log\frac{1}{\epsilon}$ oracle queries, and give a matching lower bound. Further, we implement ball optimization oracles for functions with locally stable Hessians using a variant of Newton's method. The resulting algorithm applies to a number of problems of practical and theoretical import, improving upon previous results for logistic and ℓ_∞ regression and achieving guarantees comparable to the state-of-the-art for ℓ_p regression.

1 Introduction

We study unconstrained minimization of a smooth convex objective $f : \mathbb{R}^d \to \mathbb{R}$, which we access through a *ball optimization oracle* $\mathcal{O}_{\text{ball}}$, that when queried at any point x, returns the minimizer[1] of f restricted a ball of radius r around x, i.e.,

$$\mathcal{O}_{\text{ball}}(x) = \operatorname*{arg\,min}_{x' \text{ s.t. } \|x'-x\|\le r} f(x').$$

Such oracles underlie trust region methods [12] and, as we demonstrate via applications, encapsulate problems with local stability. Iterating $x_{k+1} \leftarrow \mathcal{O}_{\text{ball}}(x_k)$ minimizes f in $\widetilde{O}(R/r)$ iterations (see Appendix A), where R is the initial distance to the minimizer, $x^\star$, and $\widetilde{O}(\cdot)$ hides polylogarithmic factors in problem parameters, including the desired accuracy.

Given the fundamental geometric nature of the ball optimization abstraction, the central question motivating our work is whether it is possible to improve upon this $\widetilde{O}(R/r)$ query complexity. It is natural to conjecture that the answer is negative: we require R/r oracle calls to observe the entire line from x_0 to the optimum, and therefore finding a solution using less queries would require jumping into completely unobserved regions. Nevertheless, we prove that the optimal query complexity scales as $(R/r)^{2/3}$. This result has positive implications for the complexity for several key regression tasks, for which we can efficiently implement the ball optimization oracles.

1.1 Our contributions

Here we overview the main contributions of our paper: accelerating ball optimization oracles (with a matching lower bound), implementing them under Hessian stability, and applying the resulting techniques to regression problems.

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†University of Washington, yintat@uw.edu.
[1] In the introduction we discuss exact oracles for simplicity, but our results account for inexactness.

arXiv:2502.02002v1 [math.OC] 4 Feb 2025

The Ball-Proximal (="Broximal") Point Method:
a New Algorithm, Convergence Theory, and Applications

Kaja Gruntkowska[1] Hanmin Li[1] Aadi Rane*[1,2] Peter Richtárik[1]

Abstract

Non-smooth and non-convex global optimization poses significant challenges across various applications, where standard gradient-based methods often struggle. We propose the *Ball-Proximal Point Method*, *Broximal Point Method*, or *Ball Point Method* (BPM) for short – a novel algorithmic framework inspired by the classical Proximal Point Method (PPM) (Rockafellar, 1976), which, as we show, sheds new light on several foundational optimization paradigms and phenomena, including non-convex and non-smooth optimization, acceleration, smoothing, adaptive stepsize selection, and trust-region methods. At the core of BPM lies the *ball-proximal ("broximal") operator*, which arises from the classical proximal operator by replacing the quadratic distance penalty by a ball constraint. Surprisingly, and in sharp contrast with the sublinear rate of PPM in the nonsmooth convex regime, we prove that BPM converges *linearly* and in a *finite* number of steps in the same regime. Furthermore, by introducing the concept of ball-convexity, we prove that BPM retains the same global convergence guarantees under weaker assumptions, making it a powerful tool for a broader class of potentially non-convex optimization problems. Just like PPM plays the role of a conceptual method inspiring the development of practically efficient algorithms and algorithmic elements, e.g., gradient descent, adaptive step sizes, acceleration (Ahn & Sra, 2020), and "W" in AdamW (Zhuang et al., 2022), we believe that BPM should be understood in the same manner: as a blueprint and inspiration for further development.

[1]King Abdullah University of Science and Technology, Thuwal, Saudi Arabia [2]University of California, Berkeley, USA.
*The work of Aadi Rane was conducted during a VSRP internship at KAUST.

1. Introduction

The minimization of non-convex functions is a fundamental challenge across many fields, including machine learning, optimization, applied mathematics, signal processing and operations research. Solving such problems is integral to most machine learning algorithms arising in both training and inference, where non-convex objectives or constraints are often necessary to capture complex prediction tasks.

1.1. Global nonconvex optimization

In this paper, we propose a new meta-algorithm (see Section 1.3) capable of finding *global* minimizers for a specific (new) class of non-convex functions. In particular, we introduce an algorithmic framework designed to solve the (potentially non-convex) optimization problem

$$\min_{x\in\mathbb{R}^d} f(x), \tag{1}$$

where $f : \mathbb{R}^d \mapsto \mathbb{R}\cup\{+\infty\}$ is assumed to be proper (which means that the set $\operatorname{dom} f := \{x \in \mathbb{R}^d : f(x) < +\infty\}$ is nonempty), closed, and have at least one minimizer. We let $\mathcal{X}_f$ be the set of all minimizers of f, and $f_\star := \min_x f(x)$.

1.2. The ball-proximal operator

A key inspiration for our method stems from the well-known Proximal Point Method (PPM) (Rockafellar, 1976), which iteratively adds a quadratic penalty term to the objective (see Section 3 for more details) and solves a modified subproblem at each step. Building on this idea, we introduce the *ball-proximal ("broximal") operator*:

Definition 1.1 (Ball-Proximal Operator)**.** *The ball-proximal ("broximal") operator* with radius $t > 0$ associated with a function $f : \mathbb{R}^d \mapsto \mathbb{R}\cup\{+\infty\}$ is given by

$$\operatorname{brox}_t^f(x) := \operatorname*{arg\,min}_{z\in B_t(x)} f(z), \tag{2}$$

where $B_t(x) := \{z \in \mathbb{R}^d : \|z - x\| \le t\}$ and $\|\cdot\|$ is the standard Euclidean norm.

According to this definition, for a given input point $x \in \mathbb{R}^d$, $\operatorname{brox}_t^f(x)$ returns the minimizer(s) of f within the ball of radius t centered at x.

arXiv:2510.00823v1 [math.OC] 1 Oct 2025

NON-EUCLIDEAN BROXIMAL POINT METHOD:
A BLUEPRINT FOR GEOMETRY-AWARE OPTIMIZATION

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ABSTRACT

The recently proposed Broximal Point Method (BPM) (Gruntkowska et al., 2025) offers an idealized optimization framework based on iteratively minimizing the objective function over norm balls centered at the current iterate. It enjoys striking global convergence guarantees, converging linearly and in a finite number of steps for proper, closed and convex functions. However, its theoretical analysis has so far been confined to the *Euclidean* geometry. At the same time, emerging trends in deep learning optimization, exemplified by algorithms such as Muon [Jordan et al., 2024] and Scion [Pethick et al., 2025], demonstrate the practical advantages of minimizing over balls defined via *non-Euclidean* norms which better align with the underlying geometry of the associated loss landscapes. In this note, we ask whether the convergence theory of BPM can be extended to this more general, non-Euclidean setting. We give a positive answer, showing that most of the elegant guarantees of the original method carry over to arbitrary norm geometries. Along the way, we clarify which properties are preserved and which necessarily break down when leaving the Euclidean realm. Our analysis positions Non-Euclidean BPM as a conceptual blueprint for understanding a broad class of geometry-aware optimization algorithms, shedding light on the principles behind their practical effectiveness.

1 Introduction

Optimization stands as a cornerstone of modern machine learning, powering the success of virtually everything, from deep neural networks to state-of-the-art large language models. As models grow ever larger and more complex, the demand for better optimization algorithms evolves in tandem. What is needed now is a new class of methods–ones that are not only computationally efficient and scalable, but also inherently robust to the nonconvex, high-dimensional landscapes that characterize contemporary deep learning.

The field of deep learning optimization has long relied on Adam and related algorithms [Kingma and Ba, 2014, Loshchilov and Hutter, 2017], whose core innovation lies in adaptive moment estimation. While these methods have achieved significant empirical success, their theoretical behavior, particularly in nonconvex landscapes, remains only partially understood. Recently, however, a new class of optimization methods has begun to challenge this long-standing dominance. Algorithms such as Muon [Jordan et al., 2024], Scion [Pethick et al., 2025] and Gluon [Riabinin et al., 2025] break away from the adaptive moment paradigm. Instead, they adopt a different principle: structured updates derived by minimizing a linear approximation of the loss function over a carefully chosen norm ball. Crucially, these norm balls are *non-Euclidean*, aiming to reflect the intrinsic geometry of the optimization problem (see Section 3.1). Operating at the individual layer level, these algorithms combine simplicity, scalability, and promising empirical performance, with several studies suggesting their potential to outperform Adam(W) in large-scale deep learning tasks [Liu et al., 2025, Pethick et al., 2025, Shah et al., 2025, Thérien et al., 2025, Yvet et al., 2025].

While Muon and Scion—both of which utilize spectral norm balls—are among the most prominent and actively studied examples in this emerging line of work, the underlying principle extends well beyond these specific instances. By changing the geometry through altering the norm, one can recover a range of familiar optimization methods. For

[1] King Abdullah University of Science and Technology

arXiv:2604.13483v1 [math.OC] 15 Apr 2026

Broximal Alignment for Global Non-Convex Optimization

Kaja Gruntkowska[1] Hanmin Li[1] Xun Qian[1] Peter Richtárik[1]

Abstract

Most non-convex optimization theory is built around gradient dynamics, leaving global convergence largely unexplored. The dominant paradigm focuses on stationarity, certifying only that the gradient norm vanishes, which is often a weak proxy for actual optimization success. In practice, gradient norms can stagnate or even increase during training, and stationary points may be far from global solutions. In this work, we propose a new framework for global non-convex optimization that avoids gradient-based reasoning altogether. We revisit the Ball Proximal Point Method (BPM), a trust-region–style algorithm introduced by Gruntkowska et al. (2025), and propose a novel structural condition – *Broximal Alignment* – under which BPM provably converges to a global minimizer. Our condition requires no convexity, smoothness, or Lipschitz assumptions, and it permits multiple and disconnected global minima as well as non-optimal local minima. We show that this class generalizes standard non-convex frameworks such as quasi-convexity, star convexity, quasar convexity, and the aiming condition. Our results provide a new conceptual foundation for global non-convex optimization beyond stationarity.

1. Introduction

Non-convex optimization lies at the heart of modern machine learning (ML). Matrix factorization (Koren et al., 2009), phase retrieval (Patterson, 1934; Candes et al., 2015), representation learning, and most prominently deep neural networks (LeCun et al., 2015; Goodfellow et al., 2016) all lead to objective functions with highly non-convex landscapes. Although such functions provide the expressive capacity needed to model complex learning tasks, they often pose significant challenges for optimizer designers. The underlying landscape can be highly irregular, with many poor local minima and saddle points that can trap optimization algorithms (Pascanu et al., 2014; Vidal et al., 2017; Safran & Shamir, 2018; Jin et al., 2021).

Unlike convex optimization, where global minimizers can be efficiently found (Nemirovski & Yudin, 1983; Nemirovski & Nesterov, 1985; Nesterov, 2003), we lack a comparable toolkit for non-convex problems. Indeed, many non-convex optimization tasks are NP-hard to solve (Murty & Kabadi, 1987), and a number of them are NP-hard even to approximate (Meka et al., 2008; Jain & Kar, 2017). As a result, rather than aiming for (approximate) global solutions, most theoretical results for non-convex optimization, especially those for gradient-based methods, are framed in terms of *stationarity*: convergence to points where the gradient norm is small. Although mathematically convenient, such guarantees provide limited insight into *global* optimization behavior.

1.1. Limitations of Stationarity-Based Non-convex Optimization Theory

To fix notation, we consider solving the general (potentially non-convex) optimization problem

$$\min_{x\in\mathbb{R}^d} f(x), \tag{1}$$

where $f : \mathbb{R}^d \mapsto \mathbb{R}\cup\{+\infty\}$ is proper (that is, the set $\operatorname{dom} f := \{x \in \mathbb{R}^d : f(x) < +\infty\}$ is nonempty), closed, and has at least one minimizer. We denote the set of all minimizers by $\mathcal{X}_f$ and write $f_\star := \min_x f(x)$.

The dominant theoretical paradigm for such tasks is built around *stationarity*. Convergence guarantees are typically stated in terms of the norm of the gradient, asserting that an algorithm reaches a point where $\|\nabla f(x)\|$ is small. This framework underlies much of the modern theory of stochastic gradient methods.

From the modeling perspective, however, stationarity-based guarantees are fundamentally misaligned with the goals of learning. A stationary point can be a global minimizer, a poor local minimizer, or a saddle point; first-order information alone does not distinguish among these possibilities. Therefore, a small gradient provides little guidance on whether an algorithm meaningfully solves the under-

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arXiv:2609.35147v1 [math.OC] 28 Sep 2026

A Sharper Theory of Ball-Proximal Optimization:
Convergence and Radius Selection

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28 September 2026

Abstract

We study the exact Euclidean ball-proximal point method for proper, closed, convex functions, where each iteration minimizes the objective over a ball centered at the current point. Retaining the objective gap in the decrease of squared distance yields sharper bounds on objective values, stationarity, and the symmetric Bregman distance to a minimizer. For constant radius $t > 0$ and initial distance $D_0 > 0$ to the solution set, the objective gap after K iterations is at most its initial value multiplied by $\exp(-2K^2t^2/D_0^2)$.

We characterize convergence for arbitrary positive radius sequences. If their sum diverges, the method reaches a minimizer in finitely many iterations whenever one exists; otherwise, the objective values converge to the infimum and the iterates escape every bounded set. If the radii are summable, the iterates converge to a possibly nonoptimal point. Self-contraction gives finite trajectory length for bounded iterates and a constant-radius termination bound $O_d(1 + D_0/t)$, whose implicit constant depends only on the dimension. A polyhedral family shows that the dimension-independent quadratic bound remains asymptotically sharp.

We also identify a minimum successful geometric decay factor and analyze adaptive radius rules based on subgradients or objective gaps, epigraph reformulation, and relaxed updates. Together, these results strengthen the foundations and convergence guarantees of the method without assuming smoothness or strong convexity.

Keywords: ball-proximal point method; convex optimization; finite termination; self-contracted sequences; radius selection; oracle complexity.
Mathematics Subject Classification (2020): Primary 90C25; Secondary 90C60, 65K05.

1 Introduction

We study the convex optimization problem

$$\min_{x\in\mathbb{R}^d} f(x), \tag{1}$$

where $f : \mathbb{R}^d \to \mathbb{R}\cup\{+\infty\}$ is proper, closed, and convex. Unless stated otherwise, we assume neither that the infimum is attained nor that f is smooth. We focus on the exact ball-oracle iteration considered by Carmon et al. (2020, Appendix A) and subsequently developed as the ball-proximal (broximal) point framework by Gruntkowska et al. (2025). Starting from $x^0 \in \operatorname{dom} f$, the ball-proximal point method (BPM) minimizes f over a Euclidean ball centered at the current point:

$$x^{k+1} \in \operatorname{brox}_t^f(x^k) := \operatorname*{arg\,min}_{u\in B_t(x^k)} f(u), \qquad t_k > 0, \tag{BPM}$$

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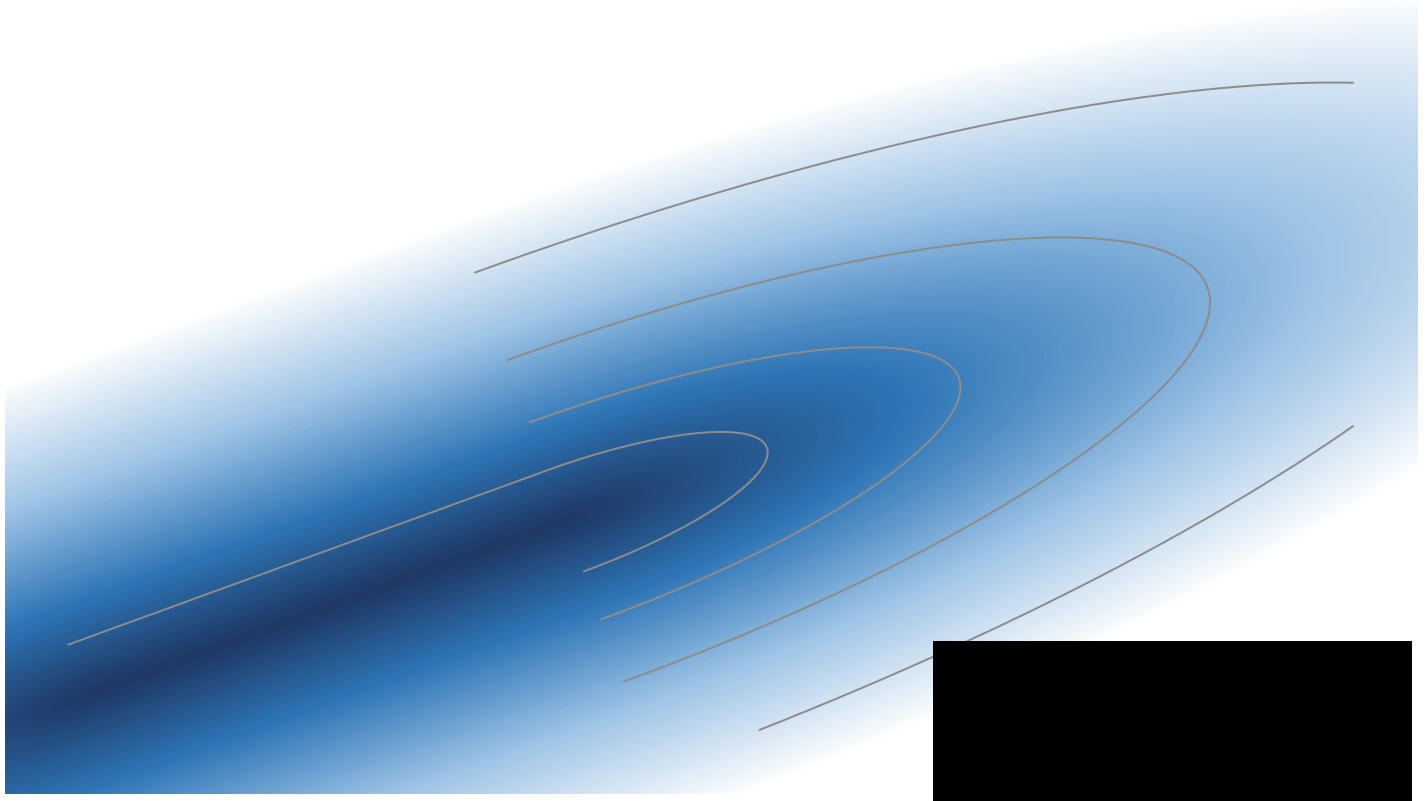
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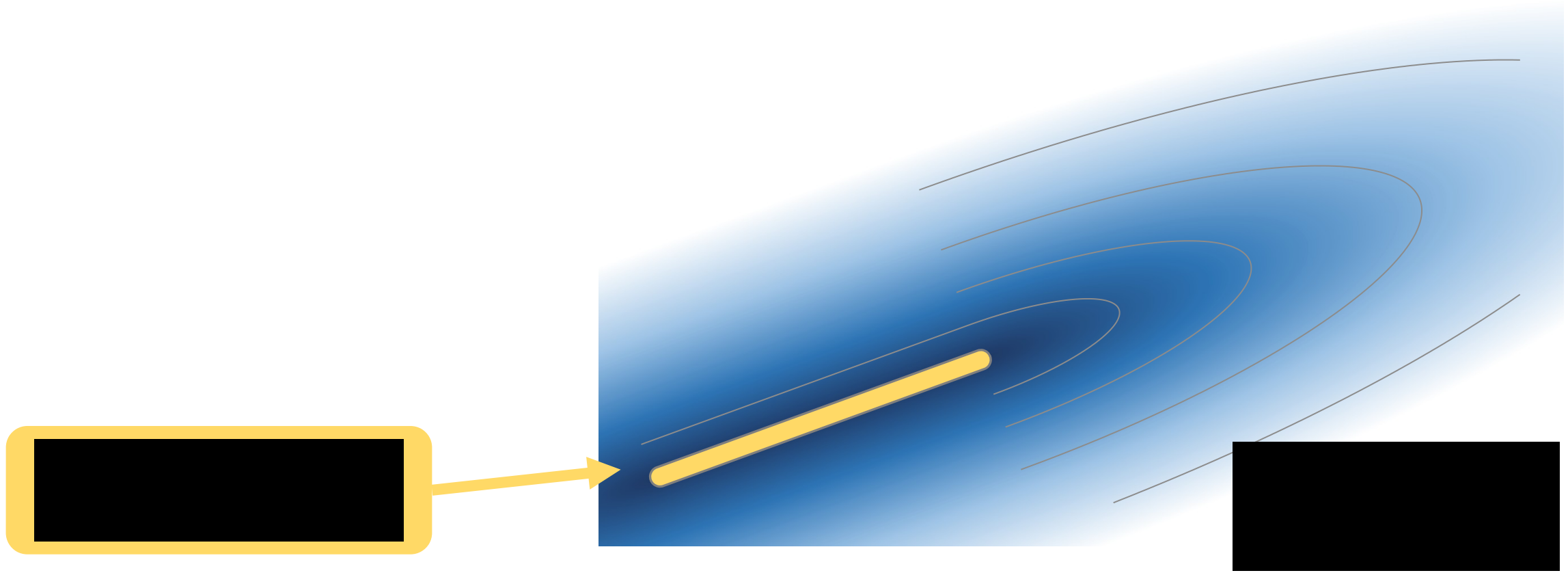


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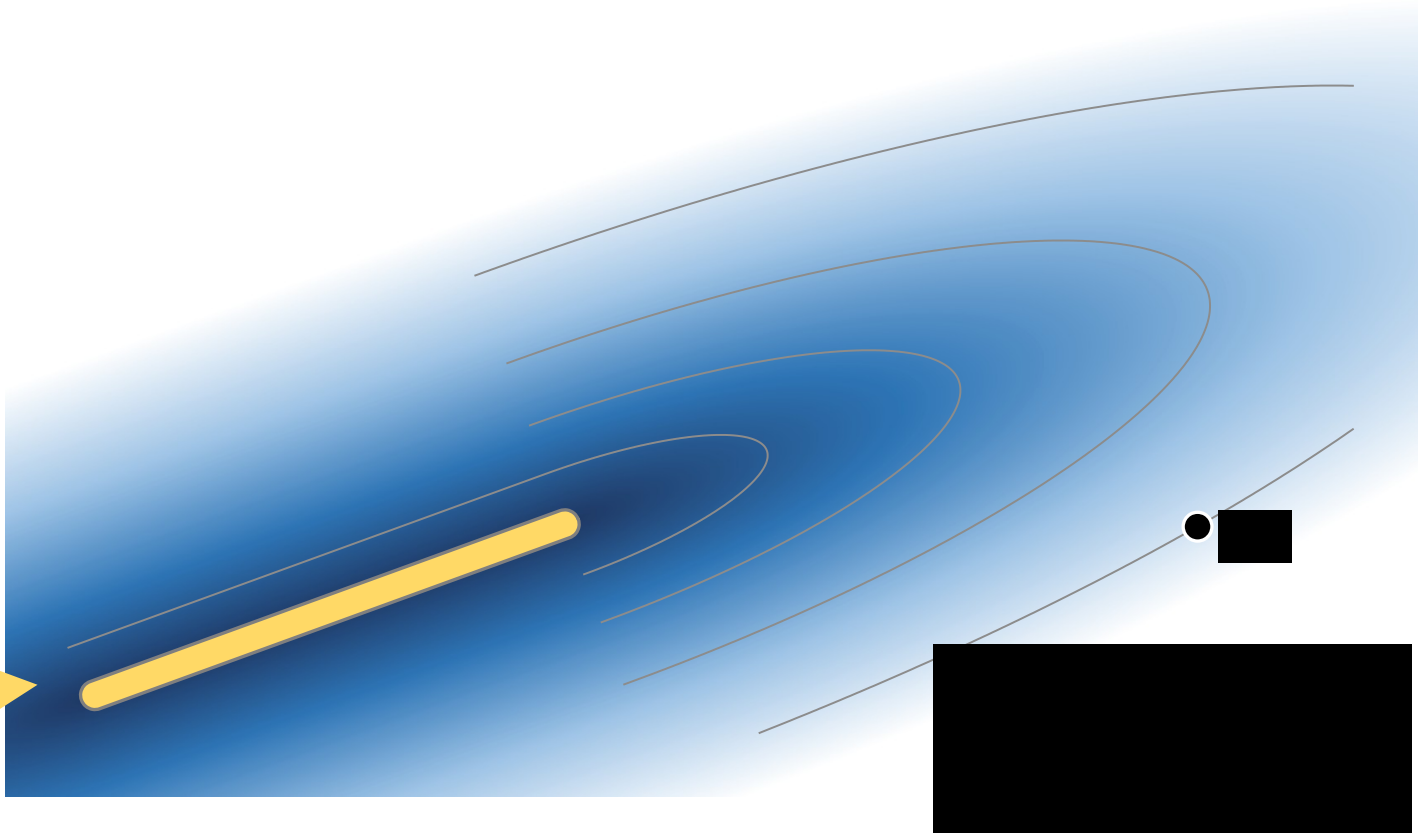
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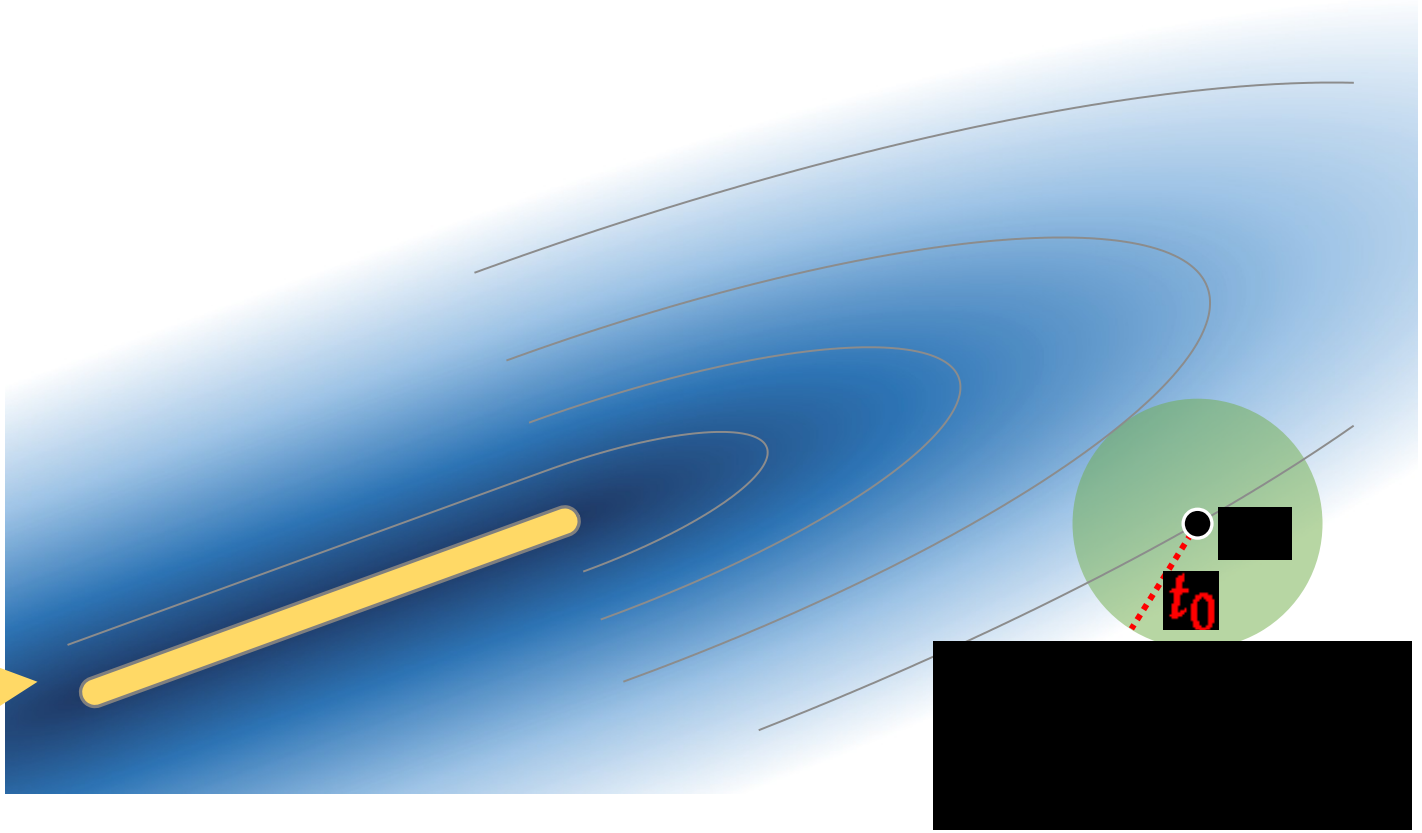
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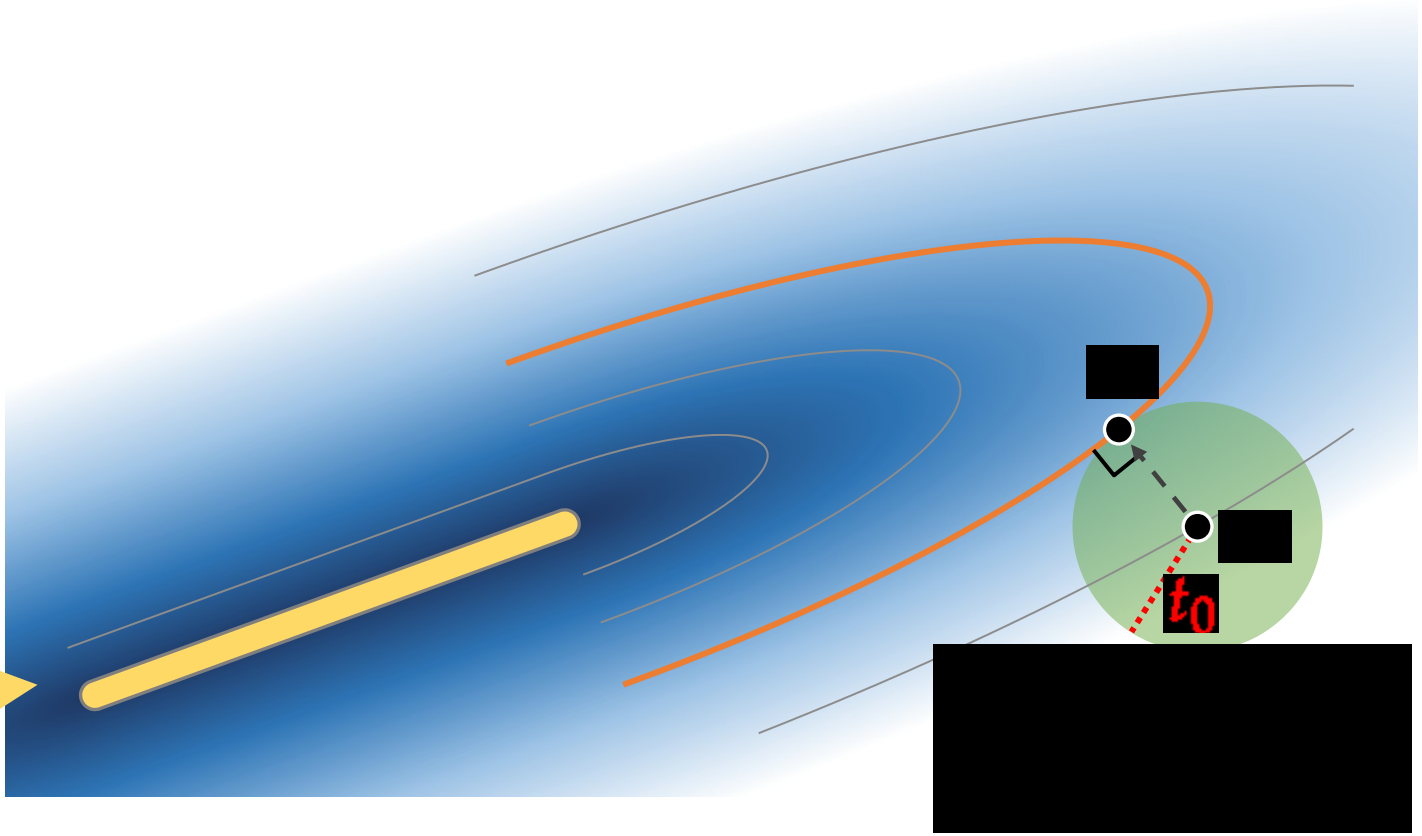


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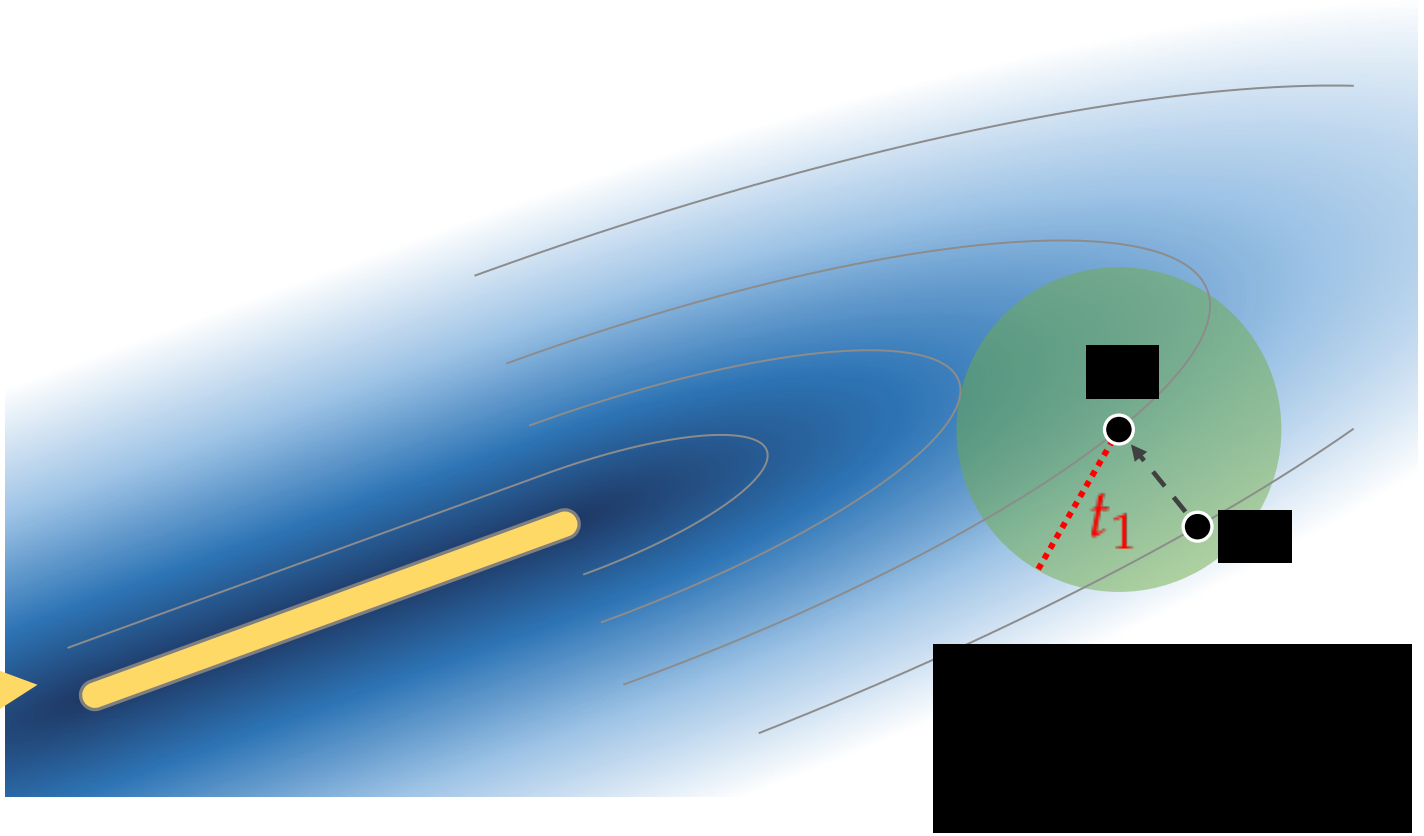


(ℓ_2 (&&

2 t_k

(

t_0



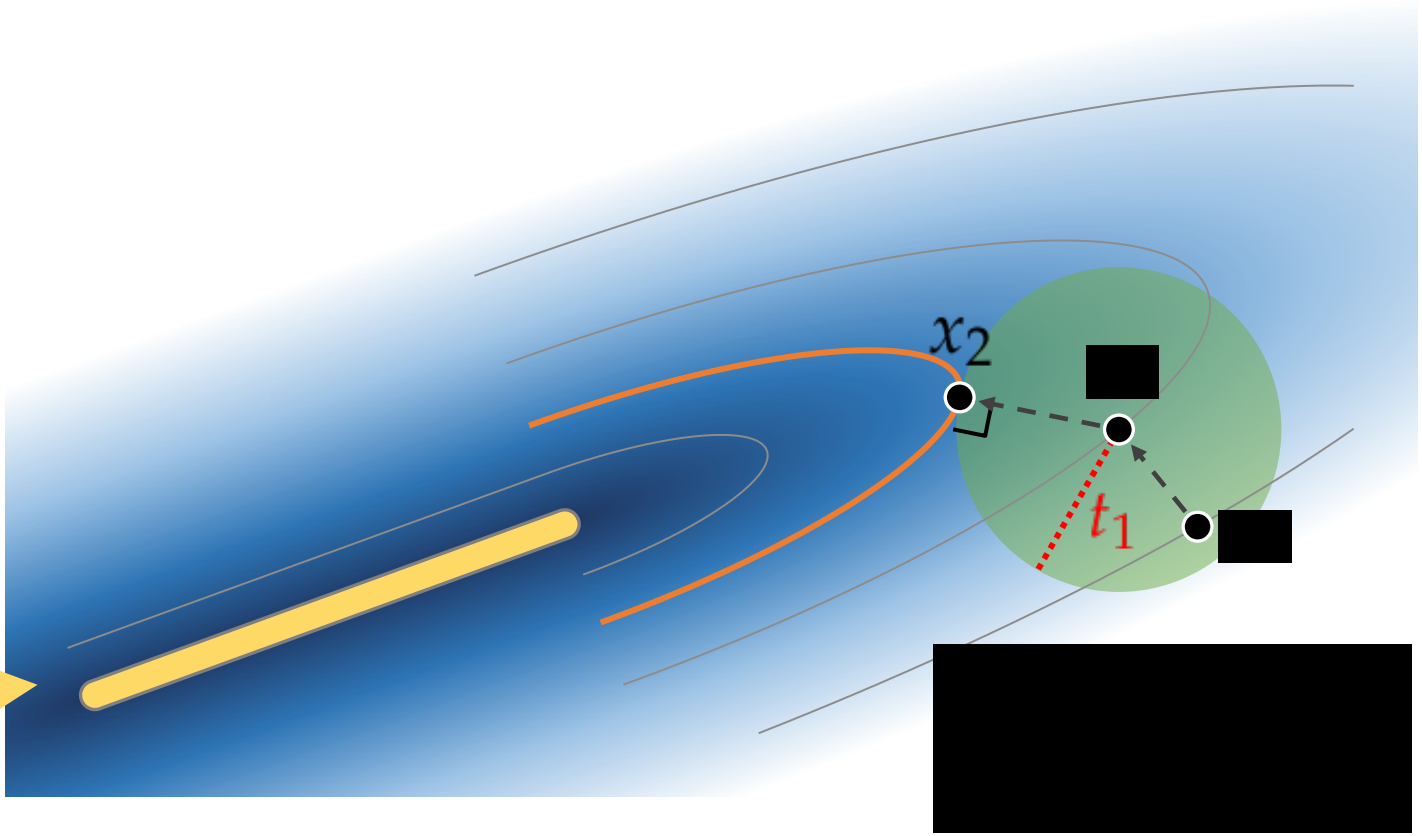
(ℓ_2 (&&

2 t_k

(

t_0

$t_1 = 1.3$ $d_2 = 3.32$ $f_2 = 5.62$ $g_2 = 3.89$



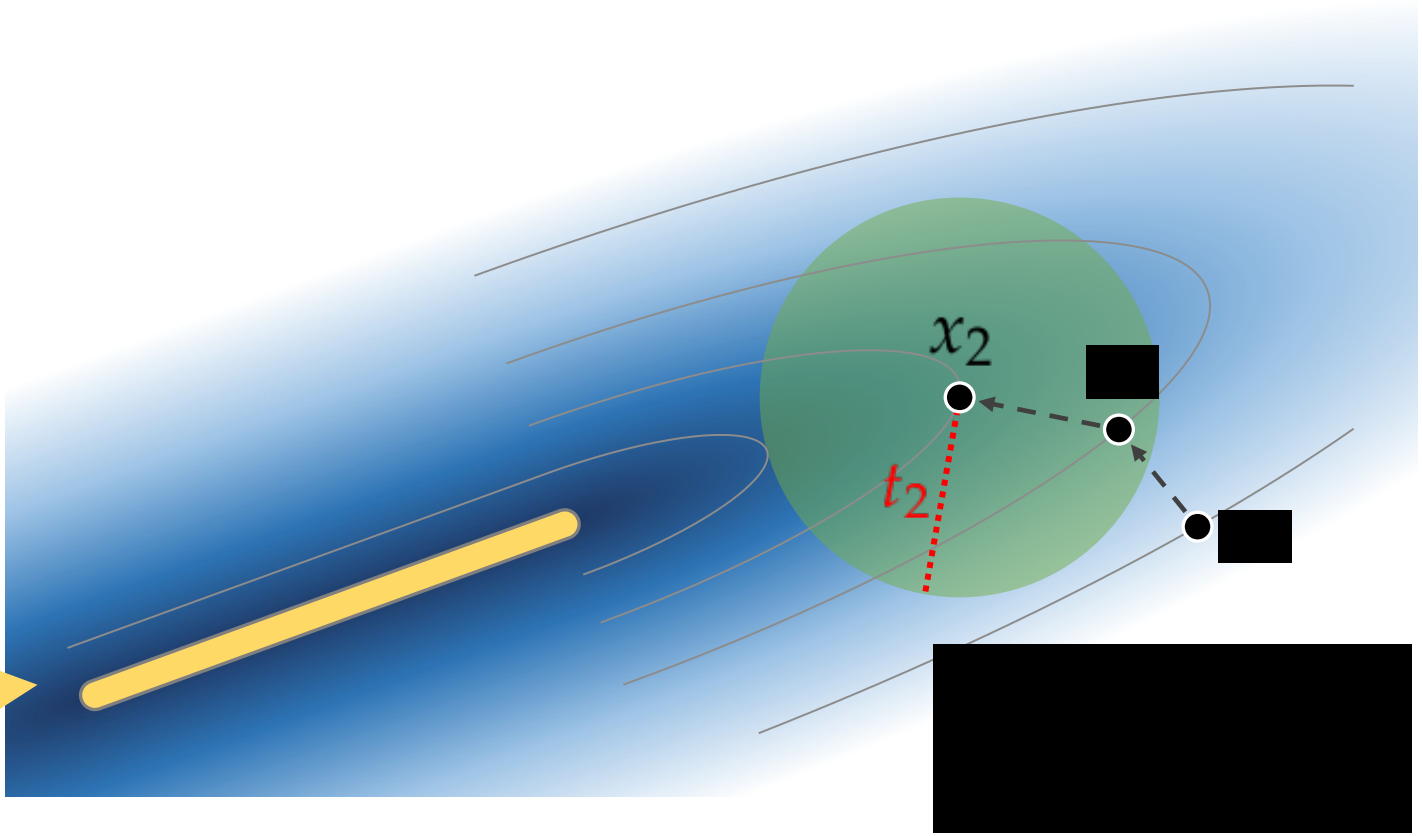
(ℓ_2 (&&

2 t_k

(

t_0

$t_1 = 1.3$ $d_2 = 3.32$ $f_2 = 5.62$ $g_2 = 3.89$



(ℓ_2 (&&

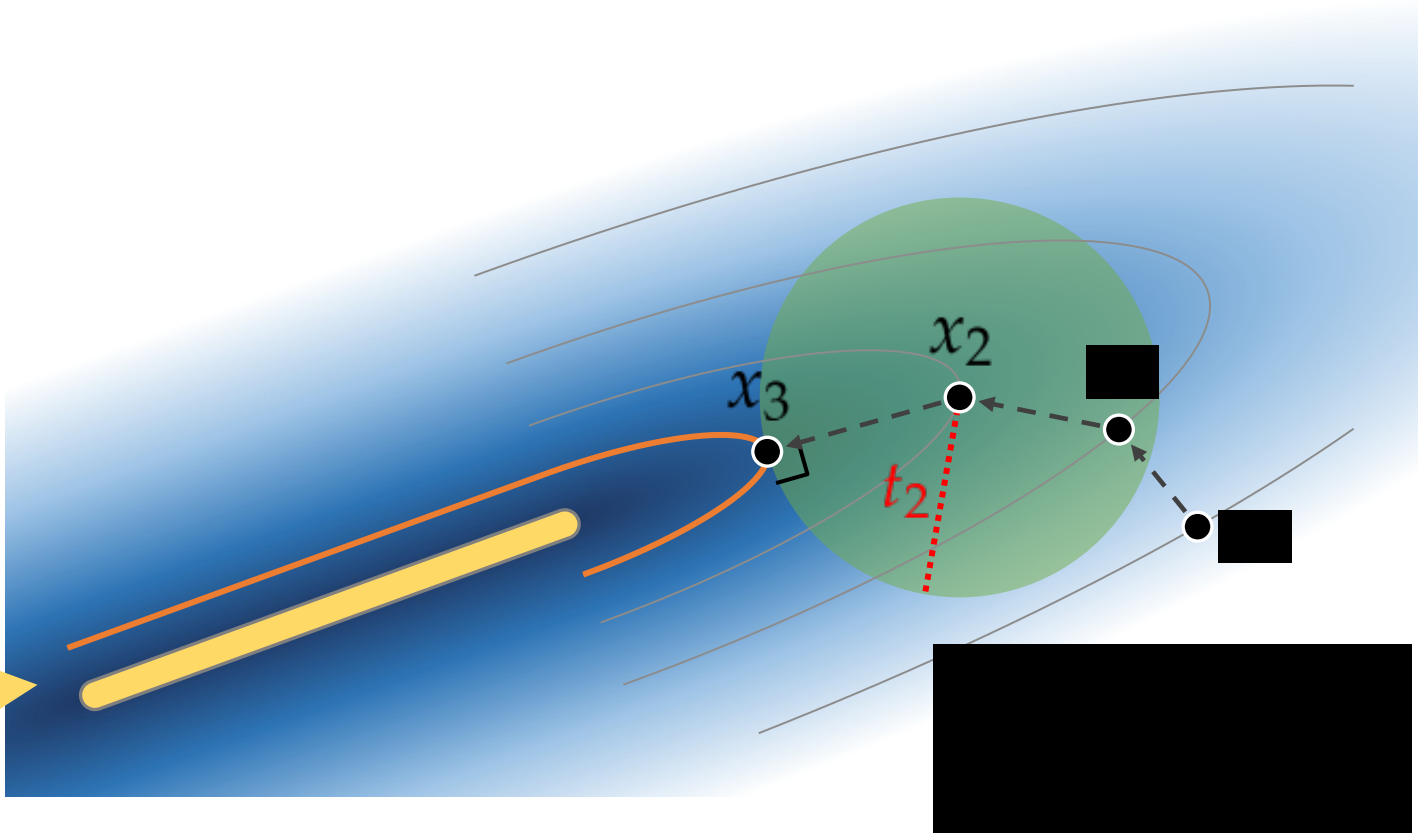
2 t_k

(

t_0

$t_1 = 1.3$ $d_2 = 3.32$ $f_2 = 5.62$ $g_2 = 3.89$

$t_2 = 1.6$ $d_3 = 1.72$ $f_3 = 1.48$ $g_3 = 1.72$

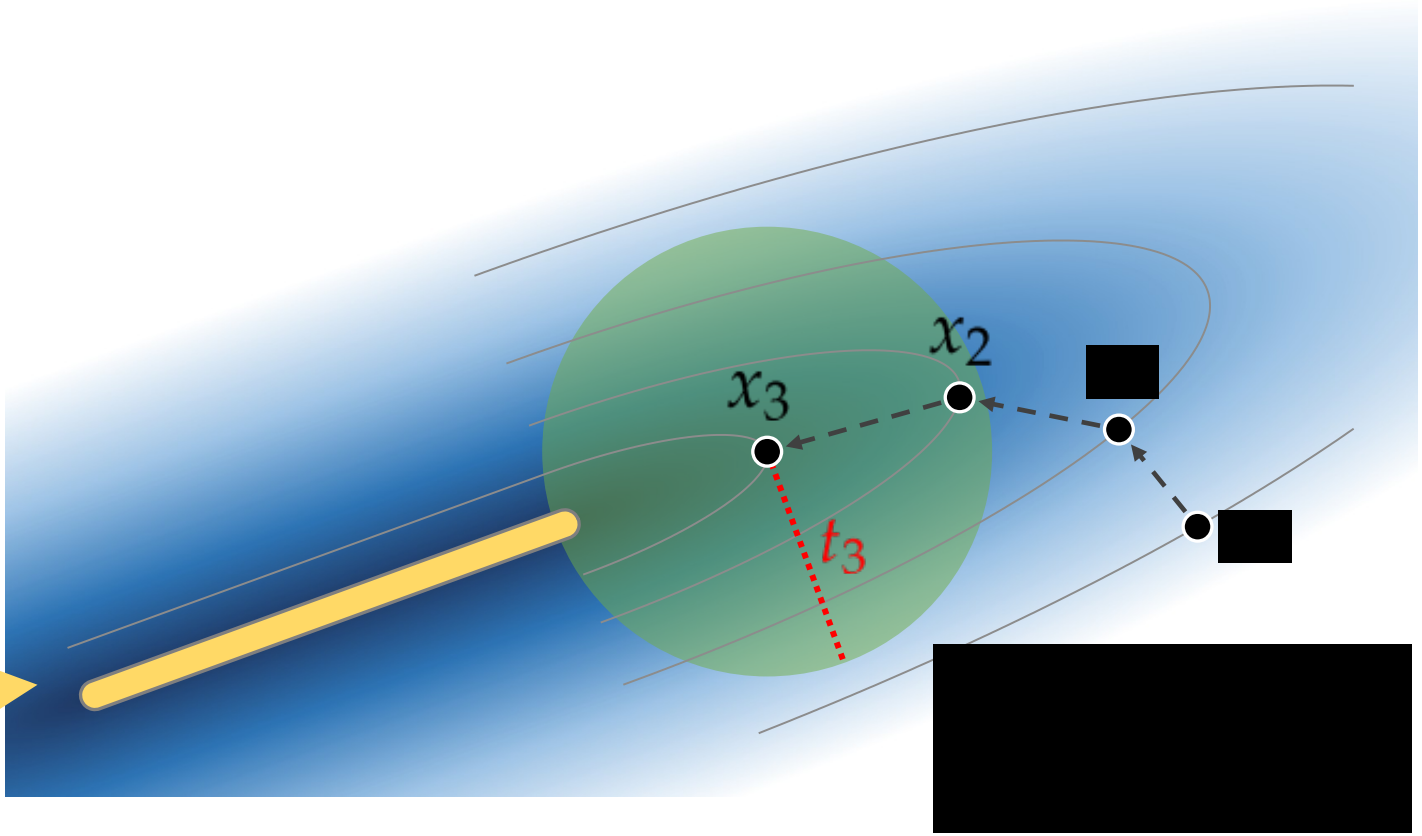


(ℓ_2 (&&

2 t_k

(

t_0			
$t_1 = 1.3$	$d_2 = 3.32$	$f_2 = 5.62$	$g_2 = 3.89$
$t_2 = 1.6$	$d_3 = 1.72$	$f_3 = 1.48$	$g_3 = 1.72$

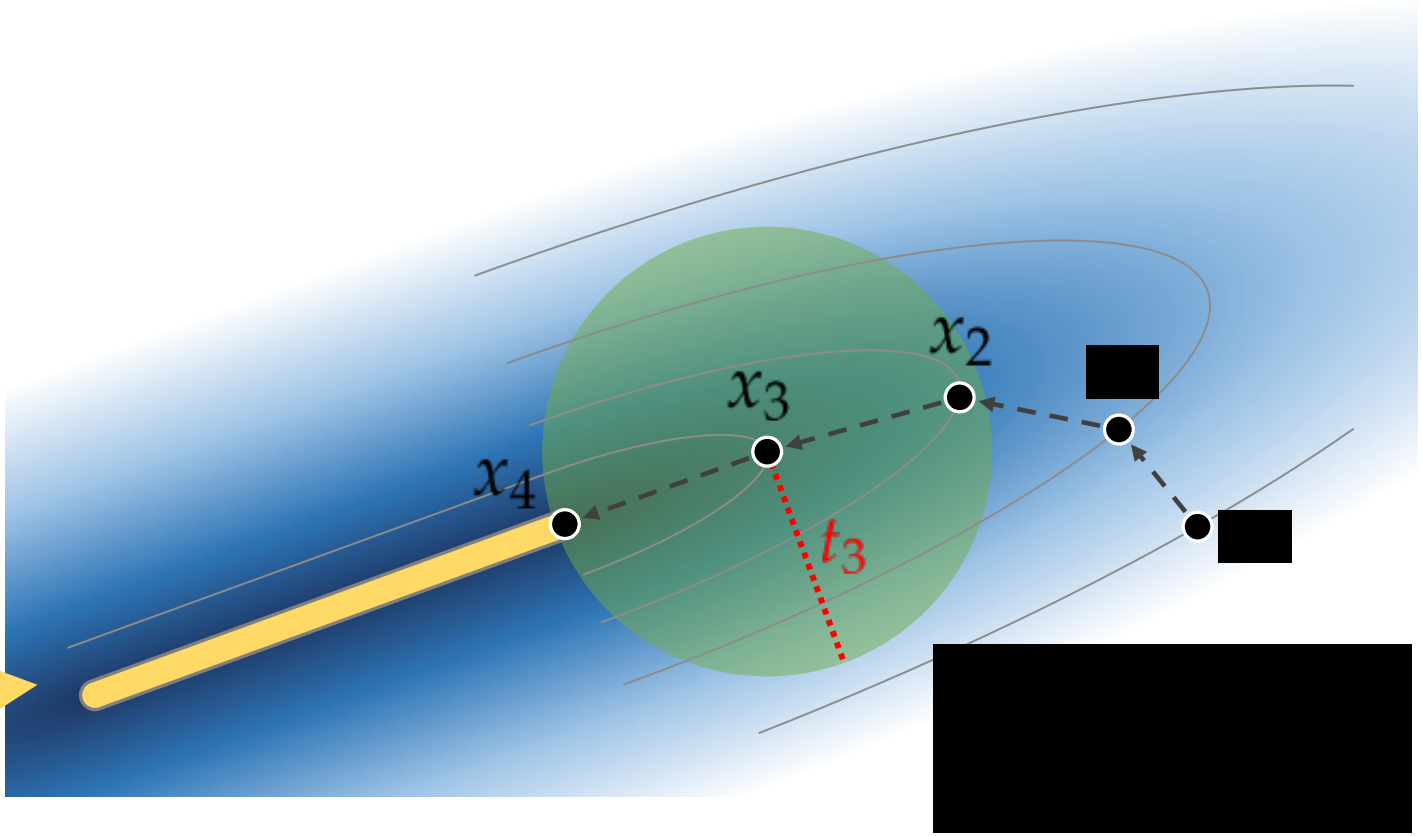


(ℓ_2 (&&

2 t_k

(

t_0			
$t_1 = 1.3$	$d_2 = 3.32$	$f_2 = 5.62$	$g_2 = 3.89$
$t_2 = 1.6$	$d_3 = 1.72$	$f_3 = 1.48$	$g_3 = 1.72$
$t_3 = 1.8$	$d_4 = 0.00$	$f_4 = 0.00$	$g_4 = 0.00$



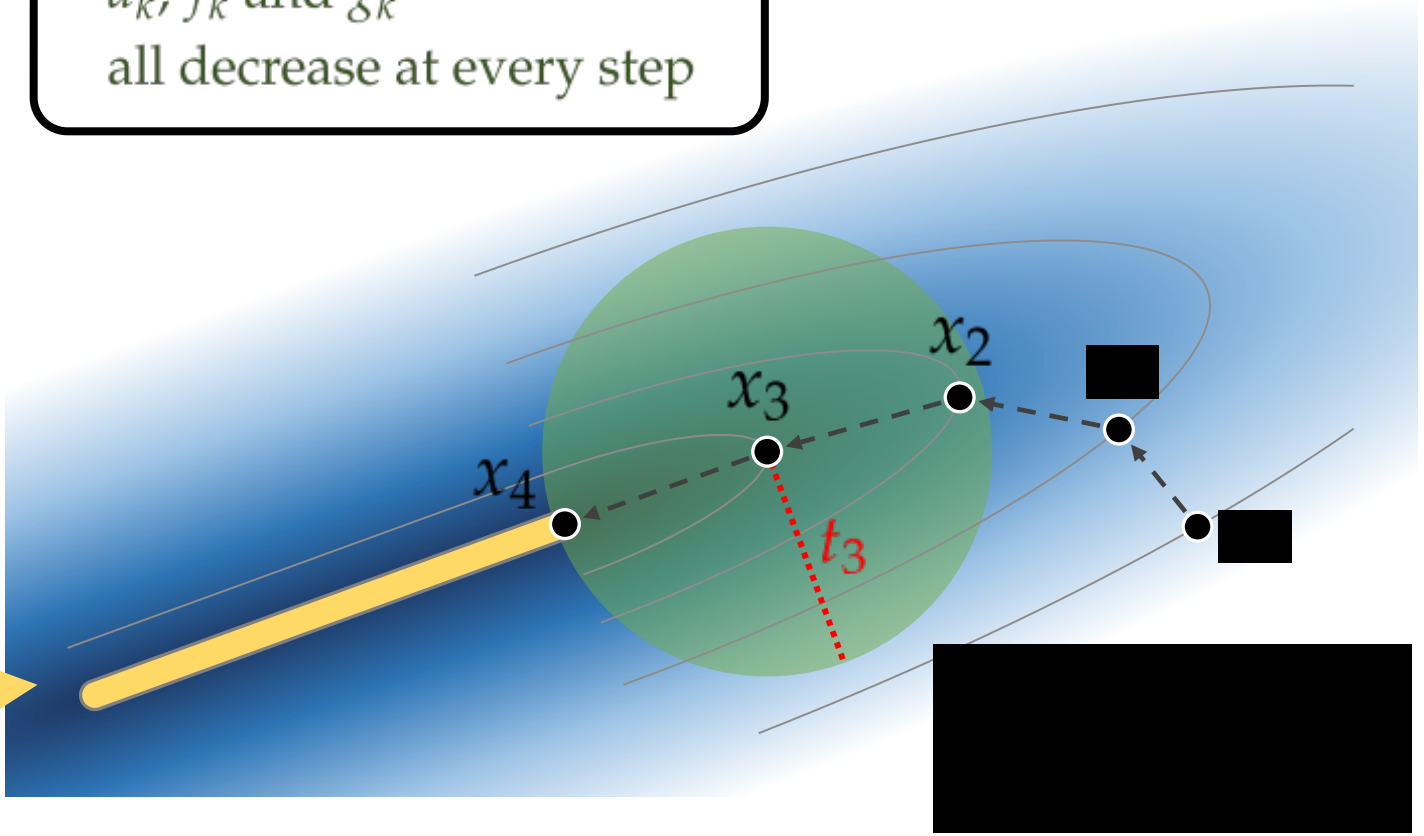
(ℓ_2 (&&

2 t_k

(

t_0			
$t_1 = 1.3$	$d_2 = 3.32$	$f_2 = 5.62$	$g_2 = 3.89$
$t_2 = 1.6$	$d_3 = 1.72$	$f_3 = 1.48$	$g_3 = 1.72$
$t_3 = 1.8$	$d_4 = 0.00$	$f_4 = 0.00$	$g_4 = 0.00$

d_k, f_k and g_k
all decrease at every step



) & ,6( ) &

*) & " 2 . 3456

$$x_{k+1} = \operatorname*{argmin}_y \left\{ f(y) + \frac{1}{2\gamma} \|y - x_k\|_2^2 \right\}$$

$$\|x_{k+1} - x_k\|_2 = \gamma \|\nabla f(x_{k+1})\|_2 \to 0$$

f differentiable: $x_k \notin \operatorname{Argmin} f$ for every k



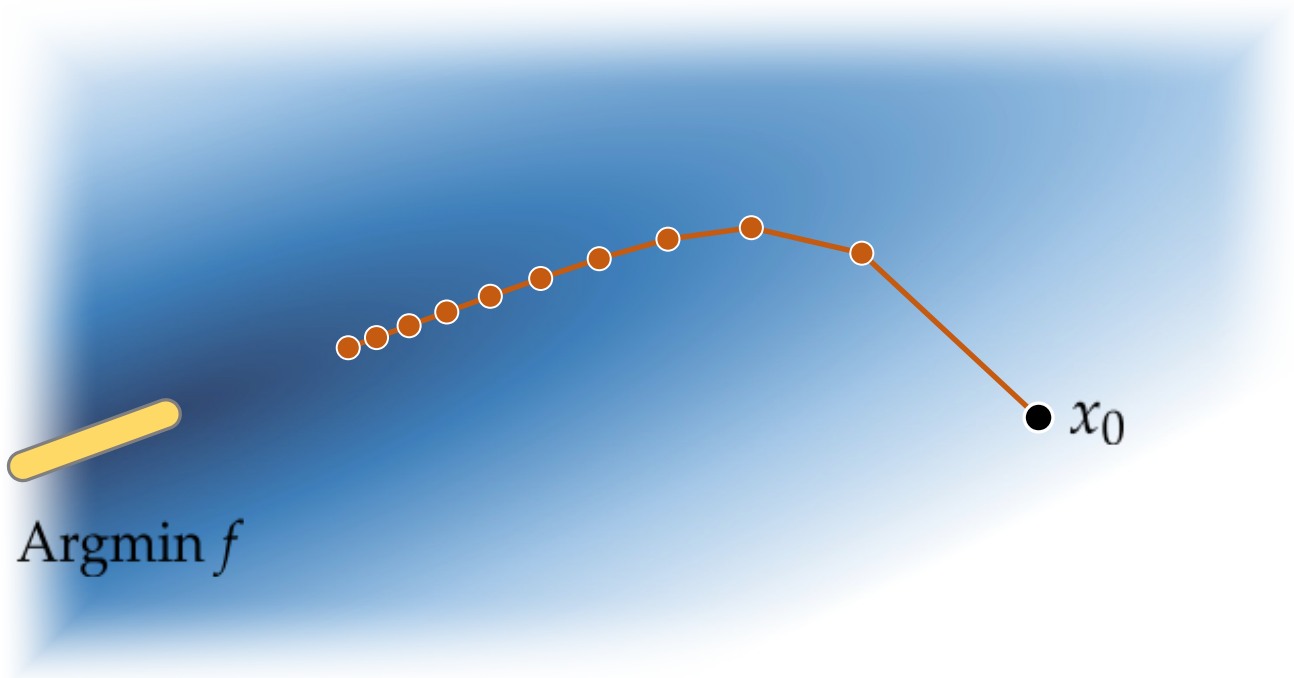
) & , 6 ( ) &

*) & " 2 . 3456

$$x_{k+1} = \operatorname*{argmin}_y \left\{ f(y) + \frac{1}{2\gamma} \|y - x_k\|_2^2 \right\}$$

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f differentiable: $x_k \notin \operatorname{Argmin} f$ for every k



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$$\|x_{k+1} - x_k\|_2 = \gamma \|\nabla f(x_{k+1})\|_2 \to 0$$

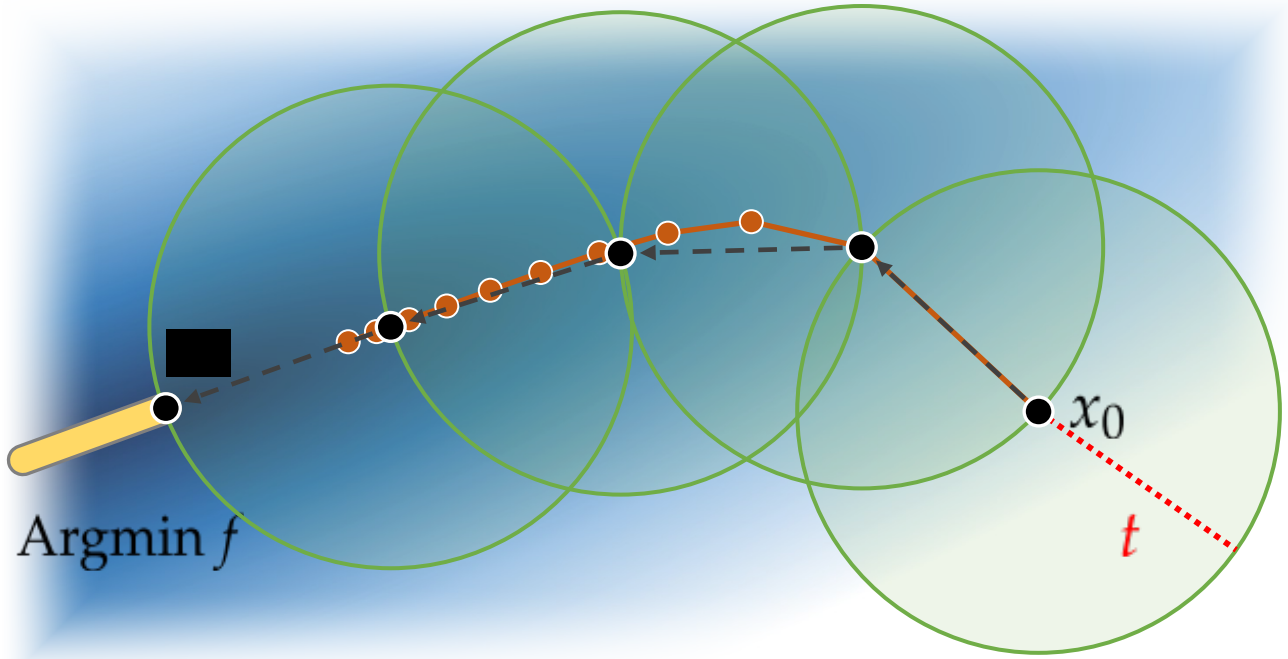
f differentiable: $x_k \notin \operatorname{Argmin} f$ for every k

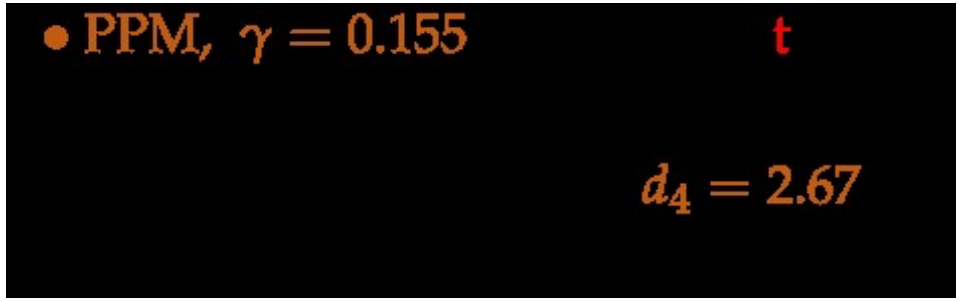
(*( ) & ./ &&-) &1 "

$$x_{k+1} \in \operatorname*{Argmin}_y \left\{ f(y) \ : \ \|y - x_k\|_2 \le t \right\}$$

$\|x_{k+1} - x_k\|_2 = t$ until the ball meets $\operatorname{Argmin} f$

exact after $\lceil d_0^2/t^2 \rceil$ steps; no condition number





) & , 6 ( ) &

*) & " 2 . 3456

$$x_{k+1} = \operatorname*{argmin}_y \left\{ f(y) + \frac{1}{2\gamma}\|y - x_k\|_2^2 \right\}$$

$$\|x_{k+1} - x_k\|_2 = \gamma \|\nabla f(x_{k+1})\|_2 \to 0$$

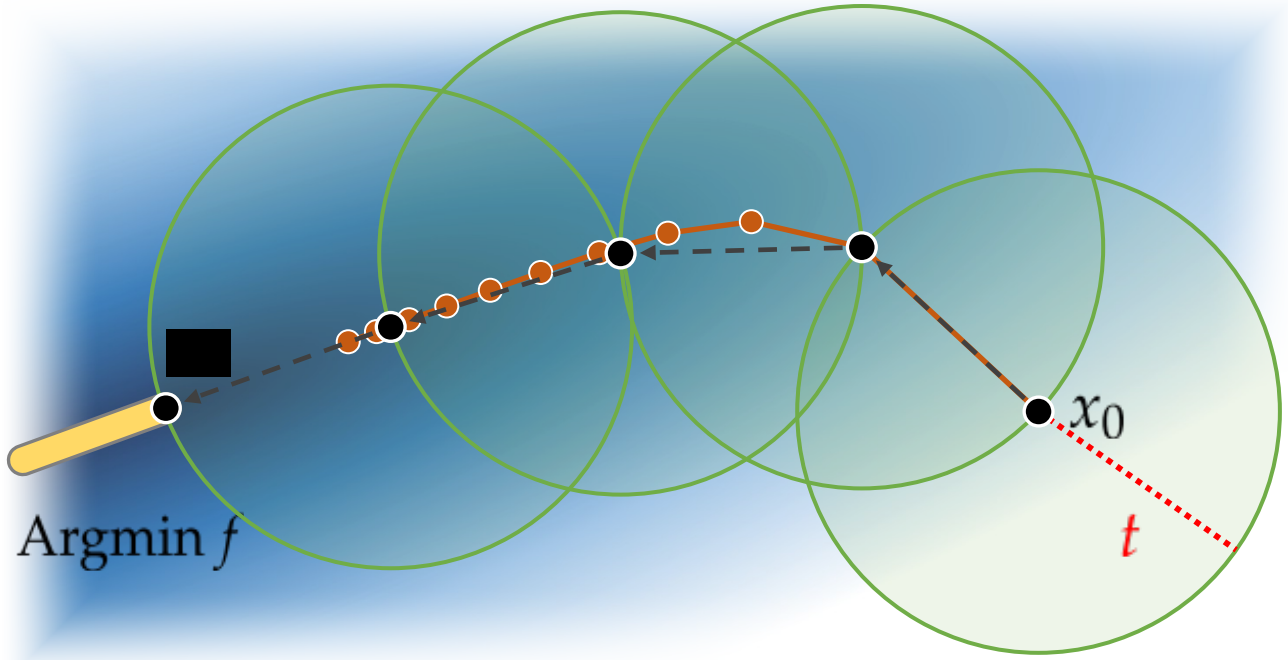
f differentiable: $x_k \notin \operatorname{Argmin} f$ for every k

(*( ) & ./ &&-) &1 "

$$x_{k+1} \in \operatorname*{Argmin}_y \left\{ f(y) \ : \ \|y - x_k\|_2 \le t \right\}$$

$\|x_{k+1} - x_k\|_2 = t$ until the ball meets $\operatorname{Argmin} f$

exact after $\lceil d_0^2/t^2 \rceil$ steps; no condition number



• PPM, $\gamma = 0.155$ t

$d_4 = 2.67$

Penalty → constraint: the steps no longer shrink, so BPM reaches Argmin f in finitely many steps.

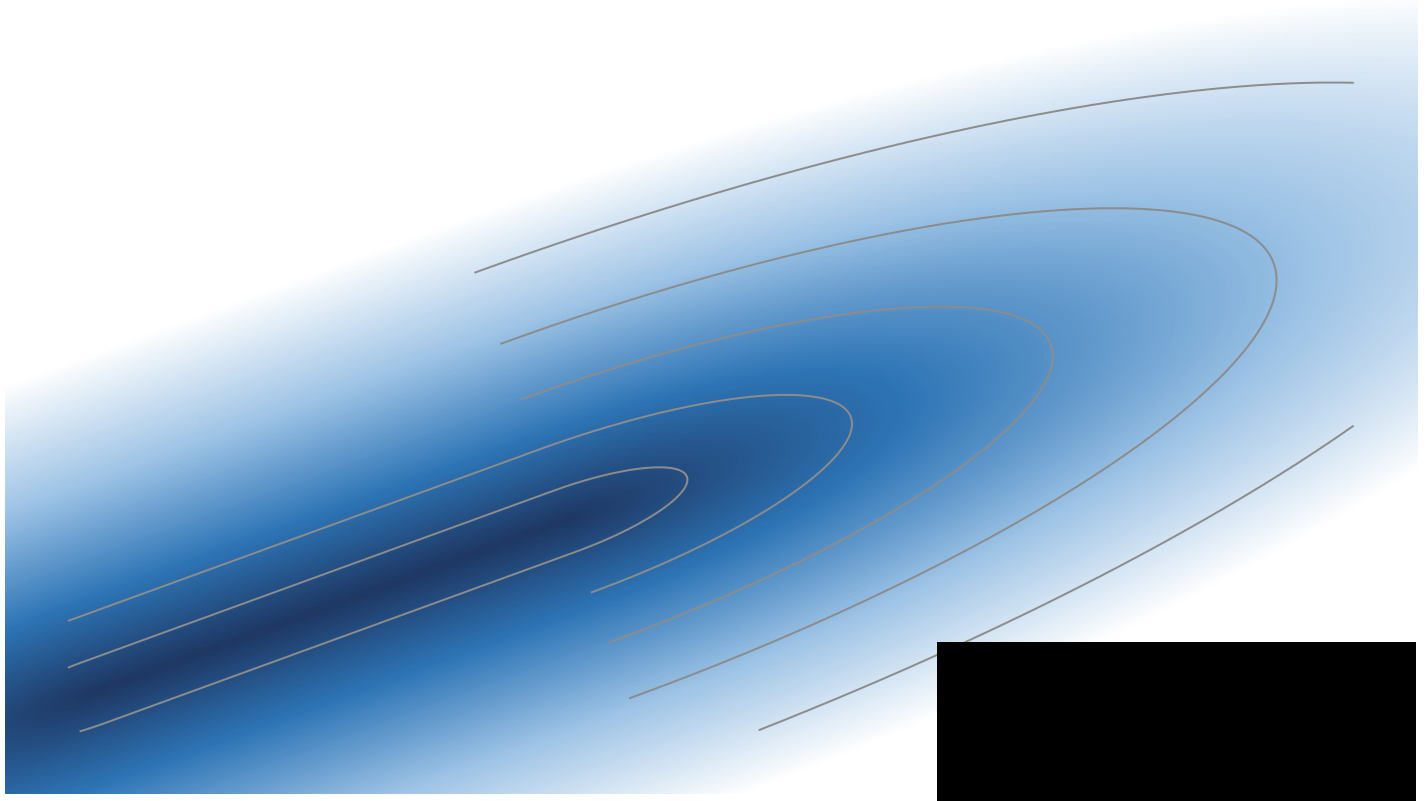
(ℓ_1 (&&





(ℓ_1 (&&

1 t_k

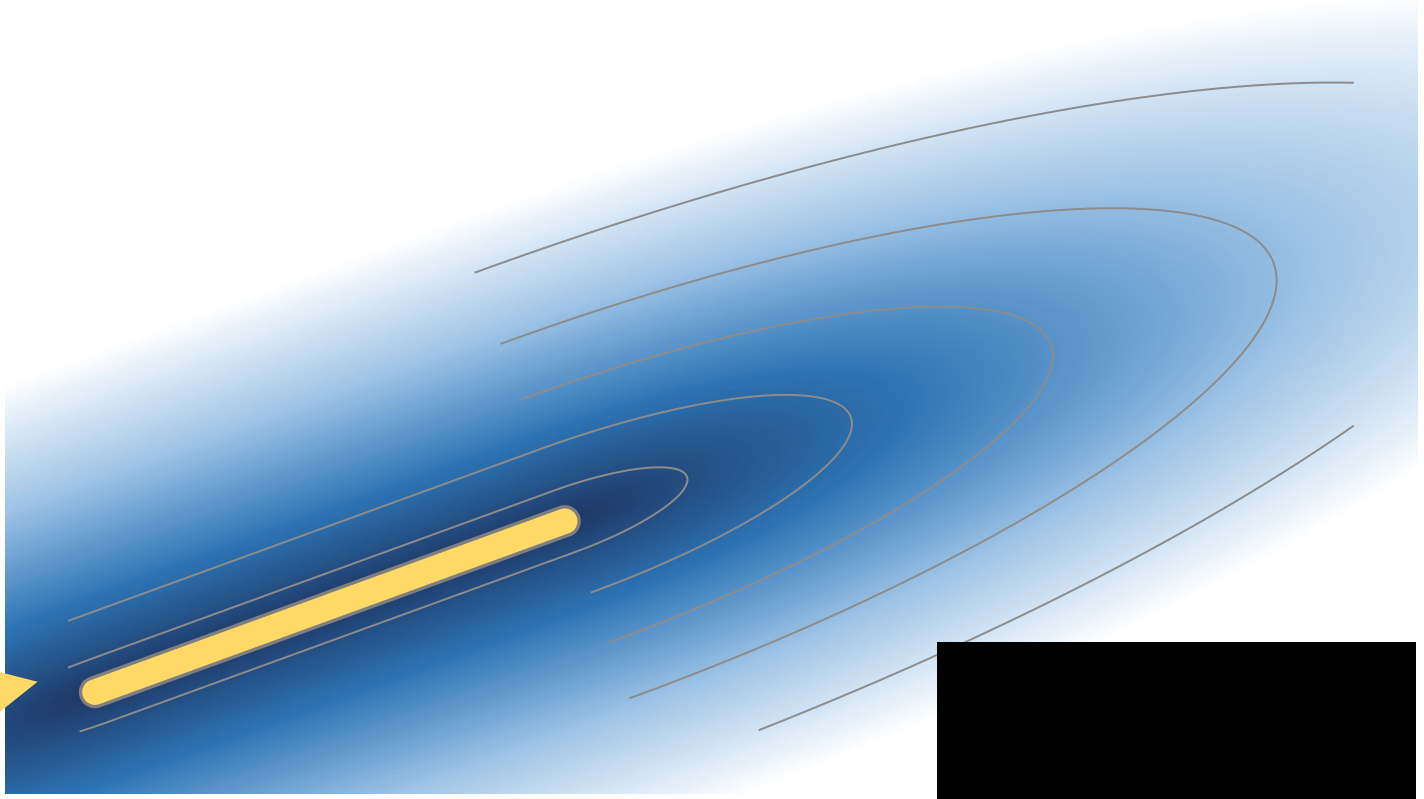


(ℓ_1 (&&

[redacted]

1 t_k

[redacted]



[redacted]

[redacted]

(ℓ_1 (&&

[redacted]

1 t_k

[redacted]

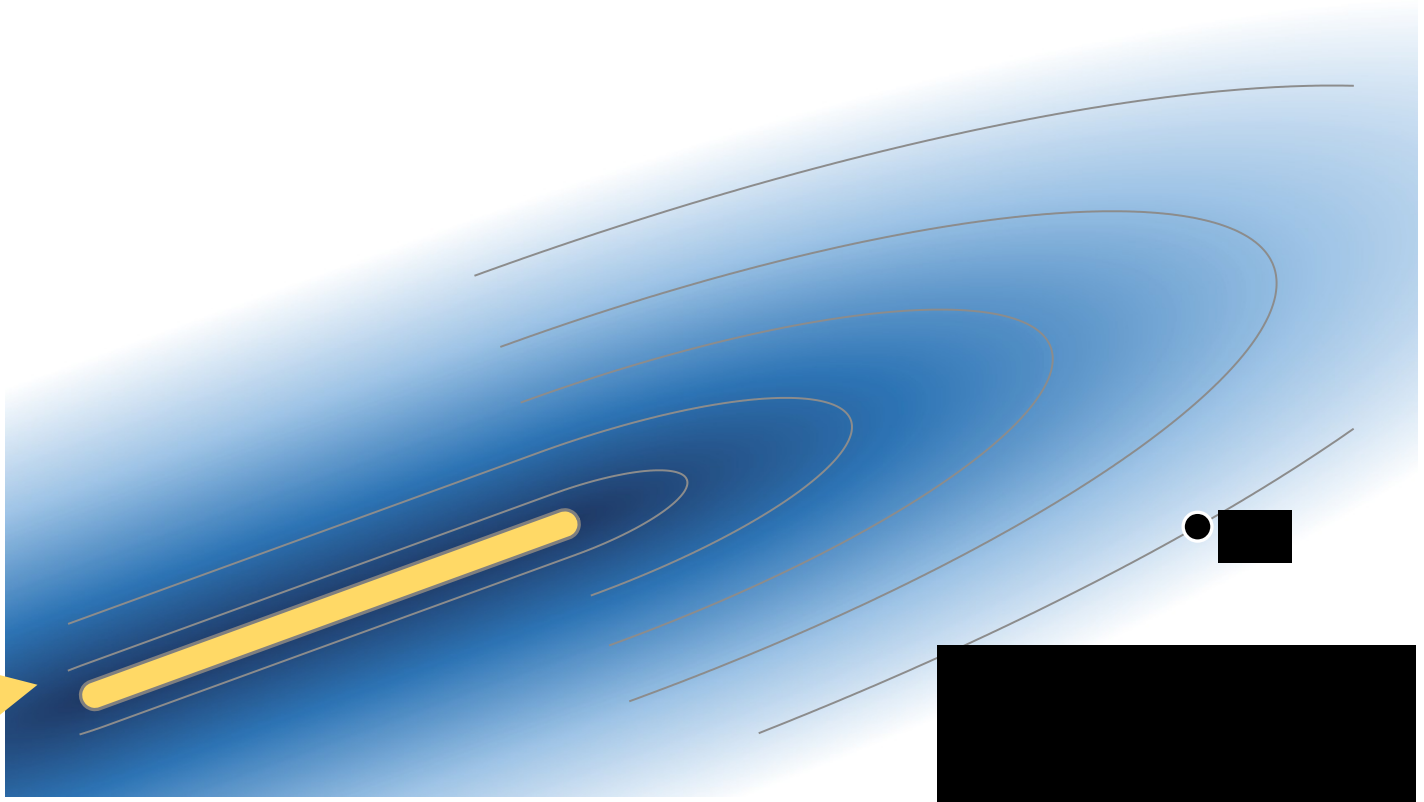
(

[redacted]

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[redacted]

(ℓ_1 (&&

[redacted]

1 t_k

[redacted]

(

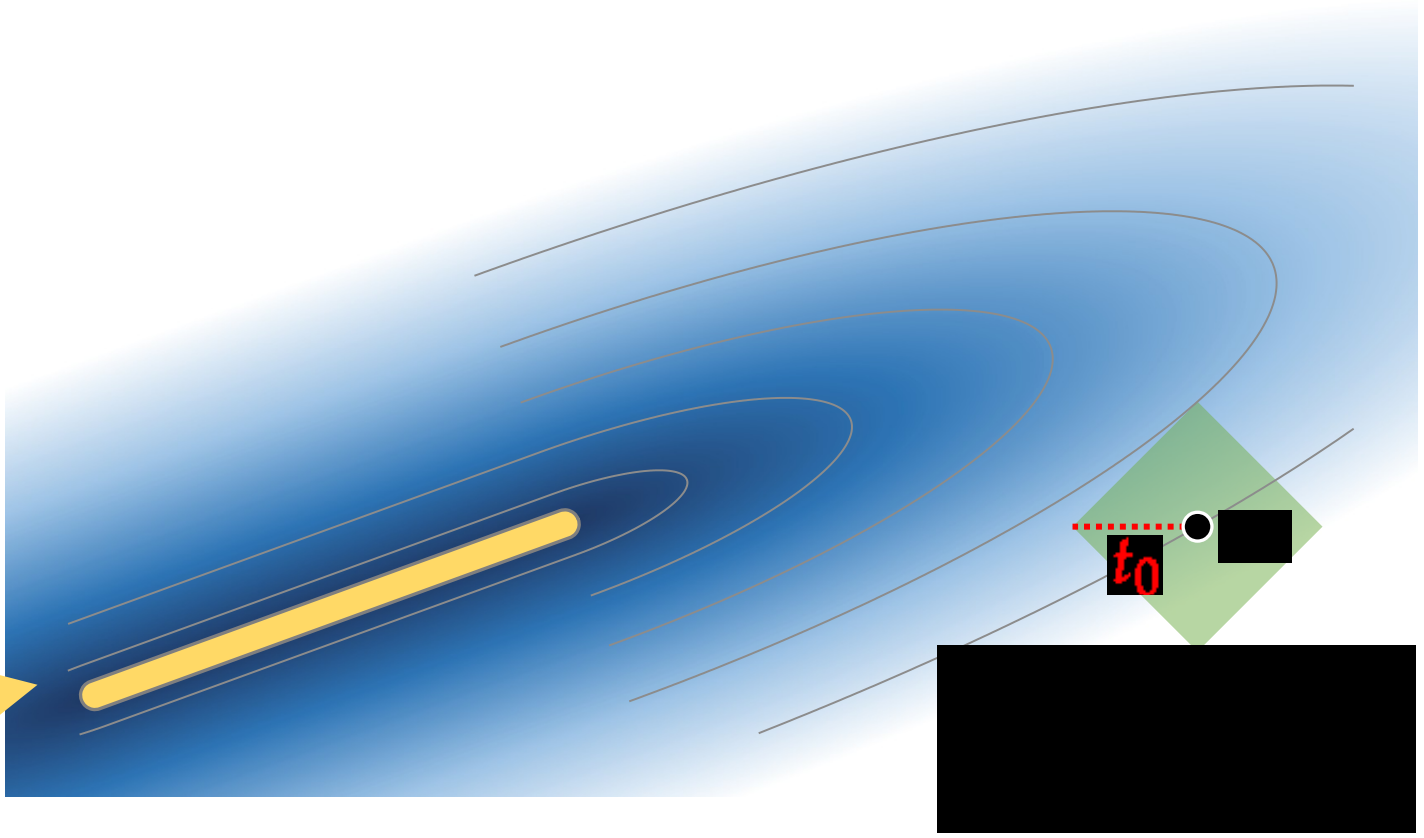
[redacted]

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(ℓ_1 (&&

[redacted]

1 t_k

[redacted]

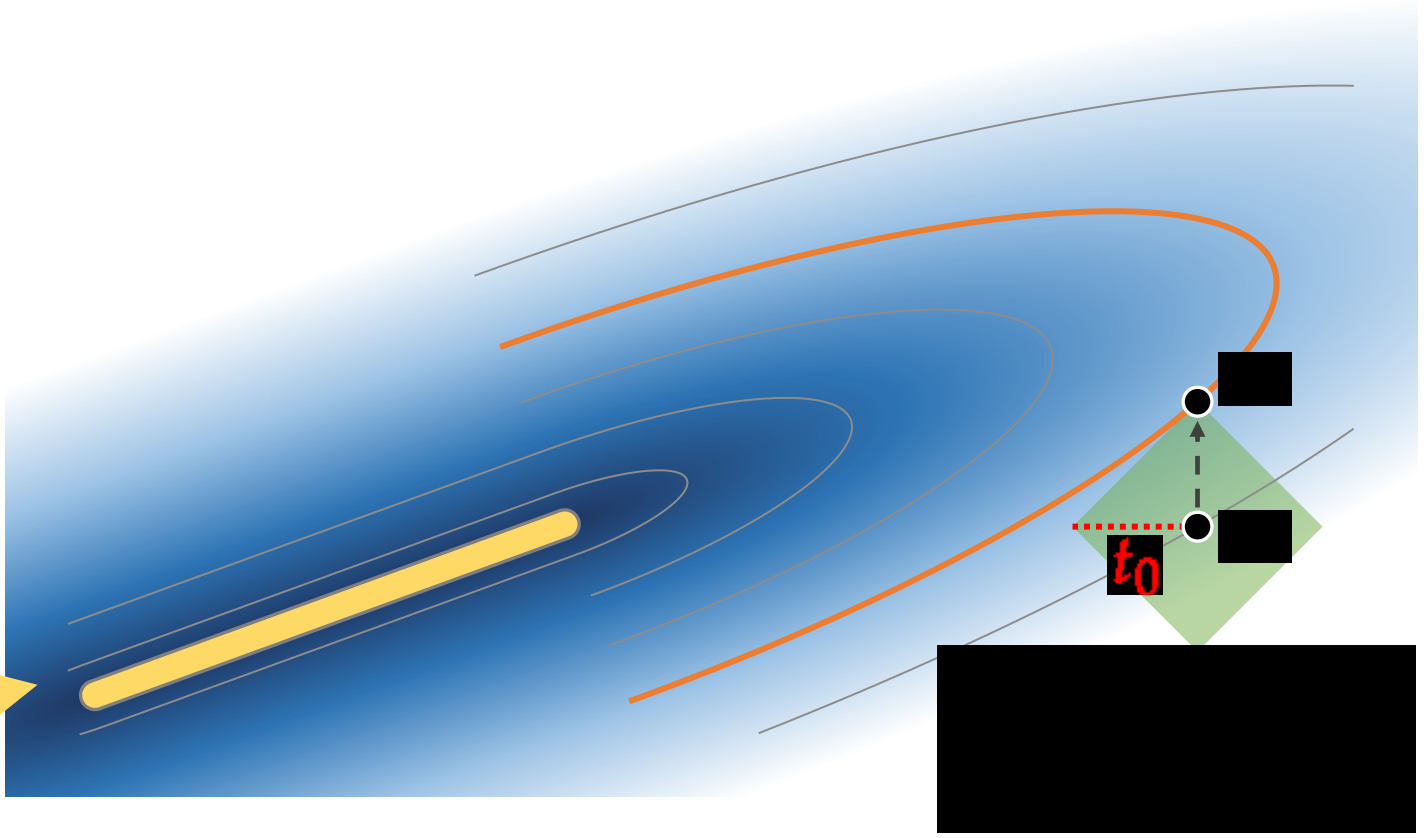
(

[redacted]

[redacted]

t_0 $d_1 = 6.04$ ↑

[redacted]



(ℓ_1 (&&

[redacted]

1 t_k

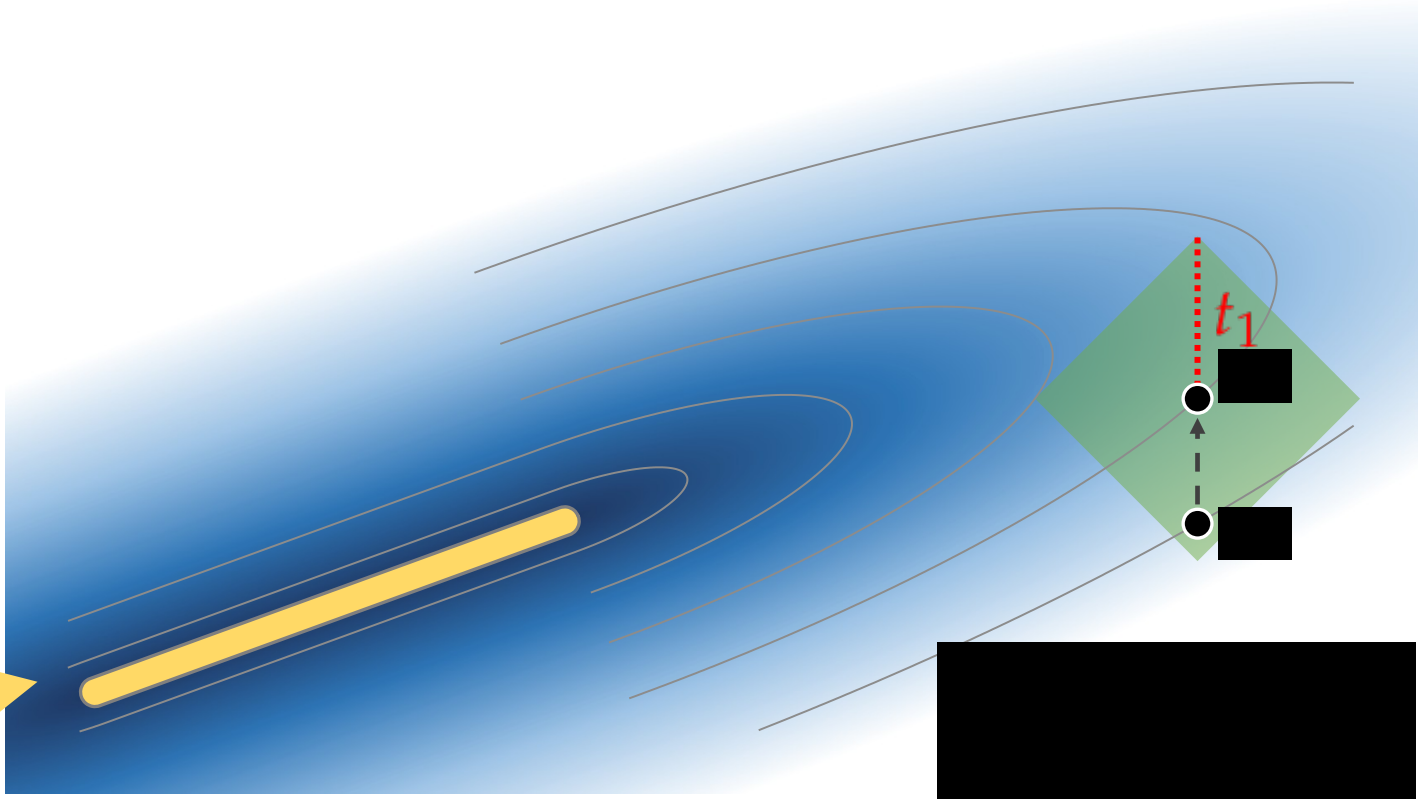
[redacted]

(

[redacted]

[redacted]

t_0 $d_1 = 6.04$ ↑



[redacted]

[redacted]

(ℓ_1 (&&

[redacted]

1 t_k

[redacted]

(

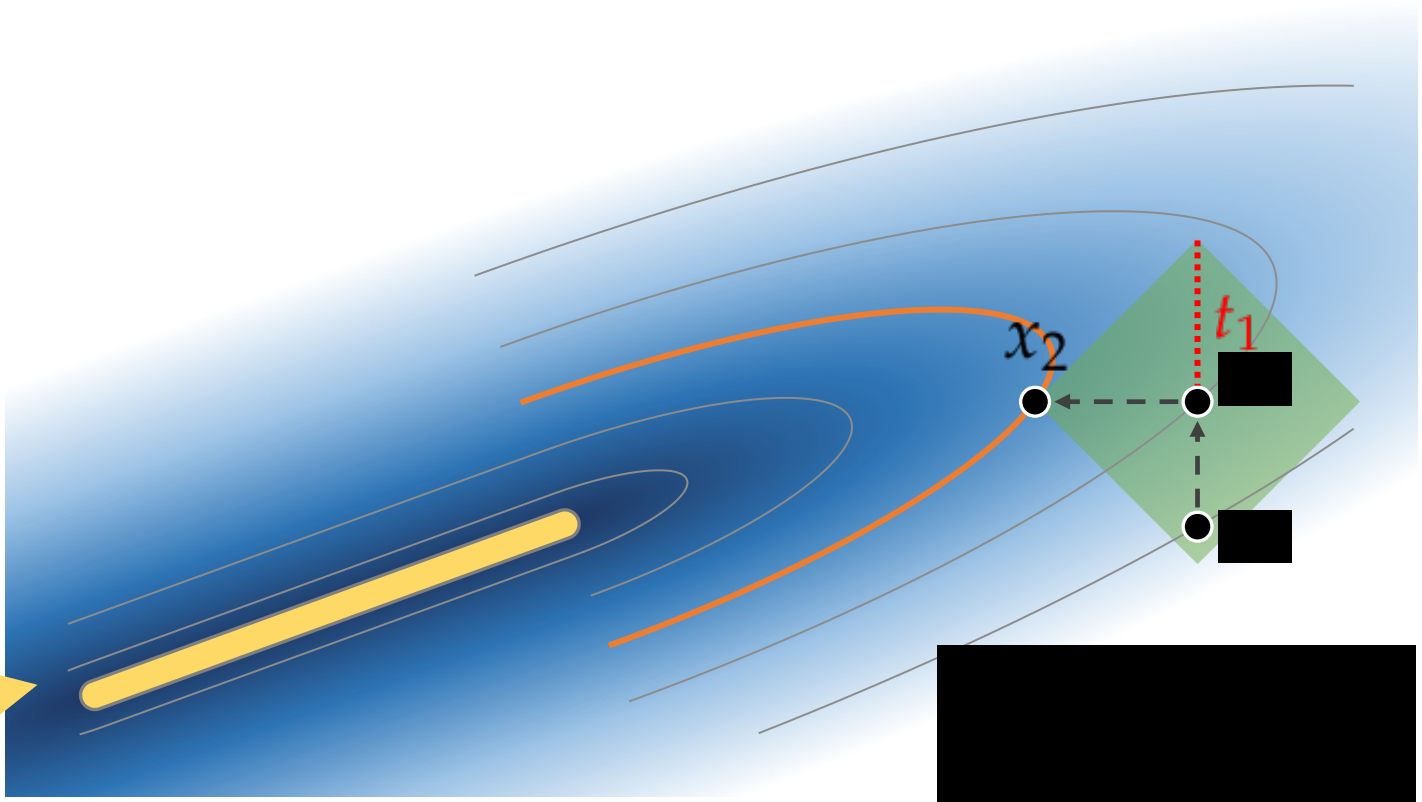
[redacted]

[redacted]

t_0 $d_1 = 6.04$ ↑

$t_1 = 1.3$

[redacted]



[redacted]

[redacted]

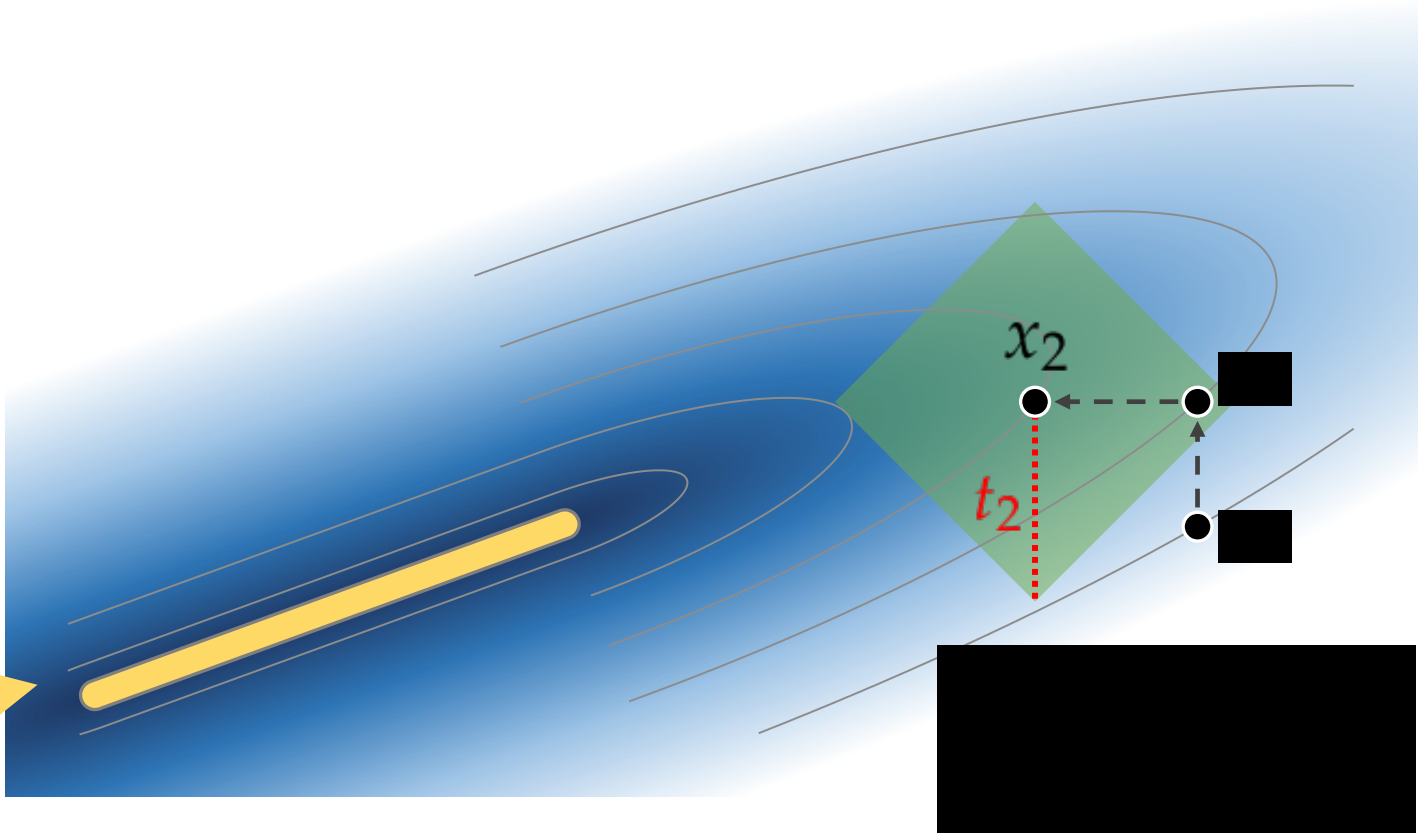
(ℓ_1 (&&

1 t_k

(

t_0 $d_1 = 6.04$ ↑

$t_1 = 1.3$



(ℓ_1 (&&

[redacted]

1 t_k

[redacted]

(

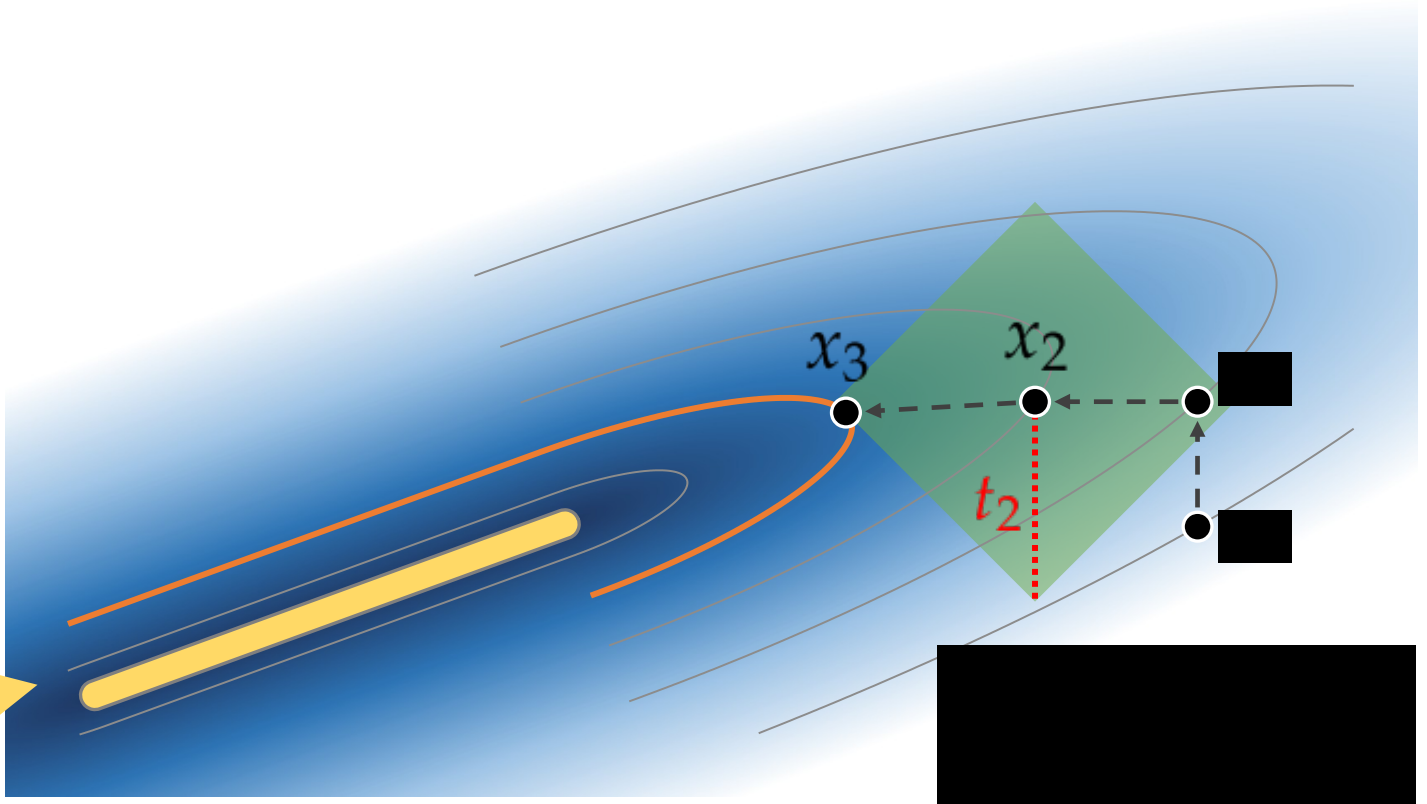
[redacted]

[redacted]

t_0 $d_1 = 6.04$ ↑

$t_1 = 1.3$

$t_2 = 1.6$



[redacted]

[redacted]

(ℓ_1 (&&

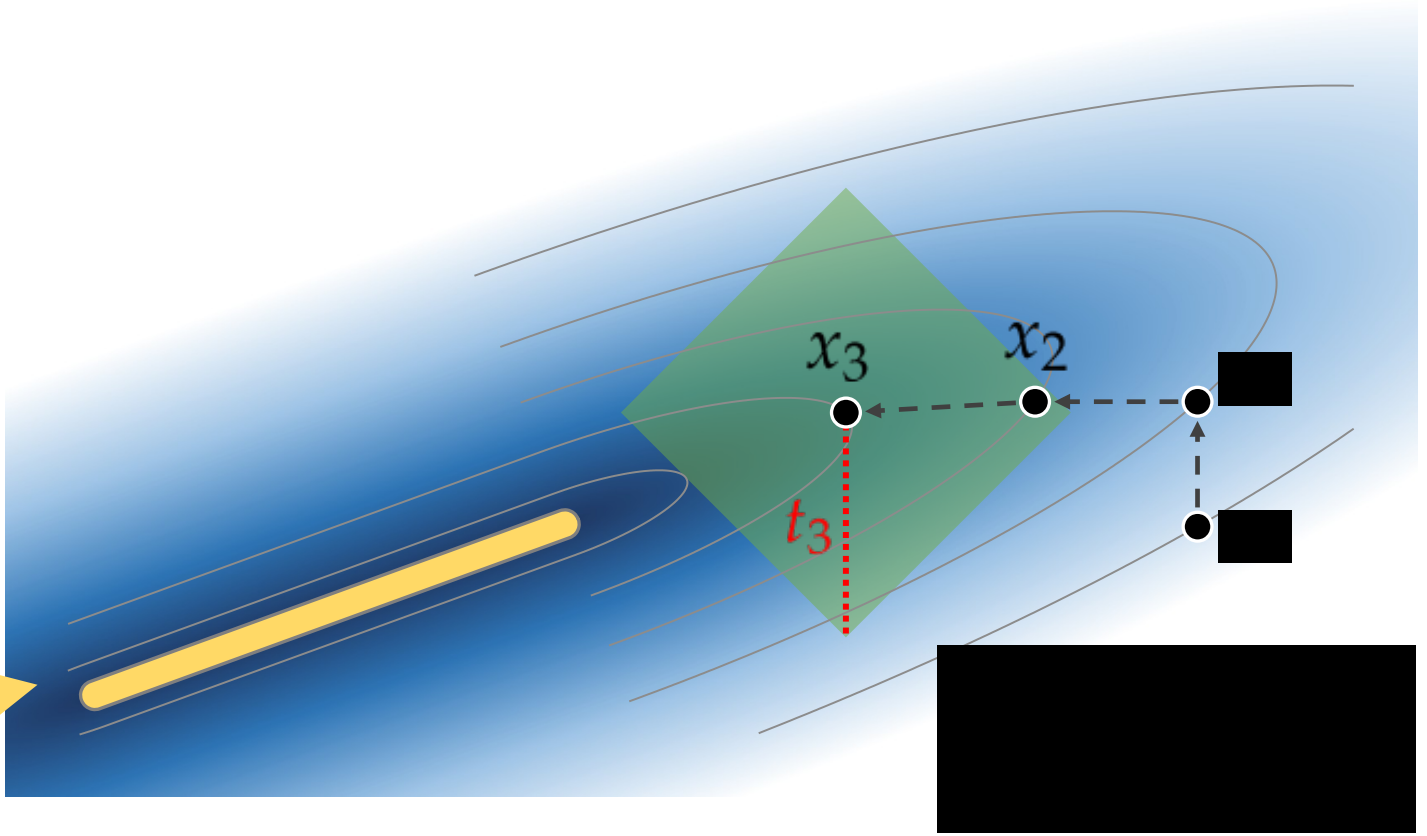
1 t_k

(

t_0 $d_1 = 6.04$ ↑

$t_1 = 1.3$

$t_2 = 1.6$



(ℓ_1 (&&

[redacted]

1 t_k

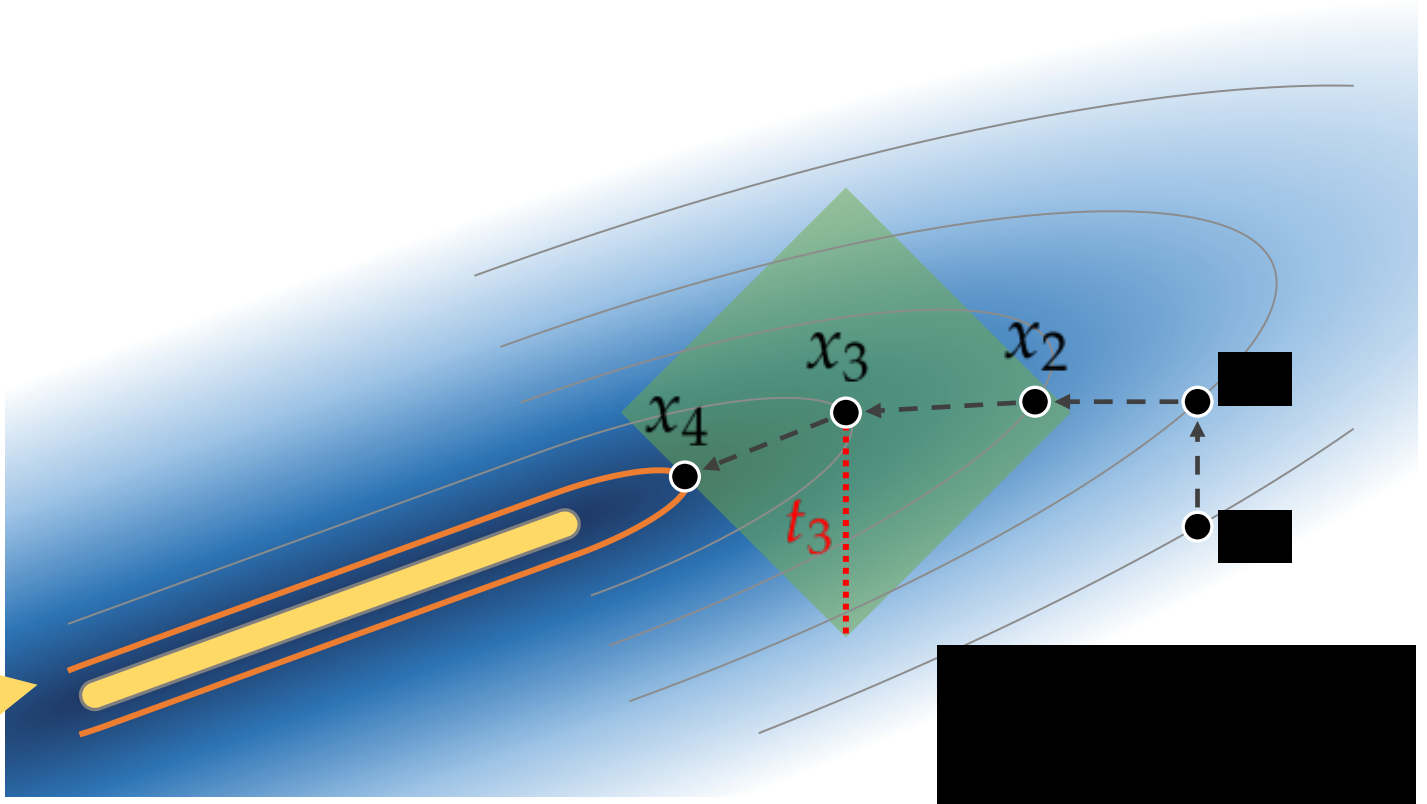
[redacted]

(

[redacted]

[redacted]			
t_0	$d_1 = 6.04$ ↑	[redacted]	[redacted]
$t_1 = 1.3$	[redacted]	[redacted]	[redacted]
$t_2 = 1.6$	[redacted]	[redacted]	[redacted]
$t_3 = 1.8$	[redacted]	[redacted]	[redacted]

[redacted]



[redacted]

(ℓ_1 (&&

[redacted]

1 t_k

[redacted]

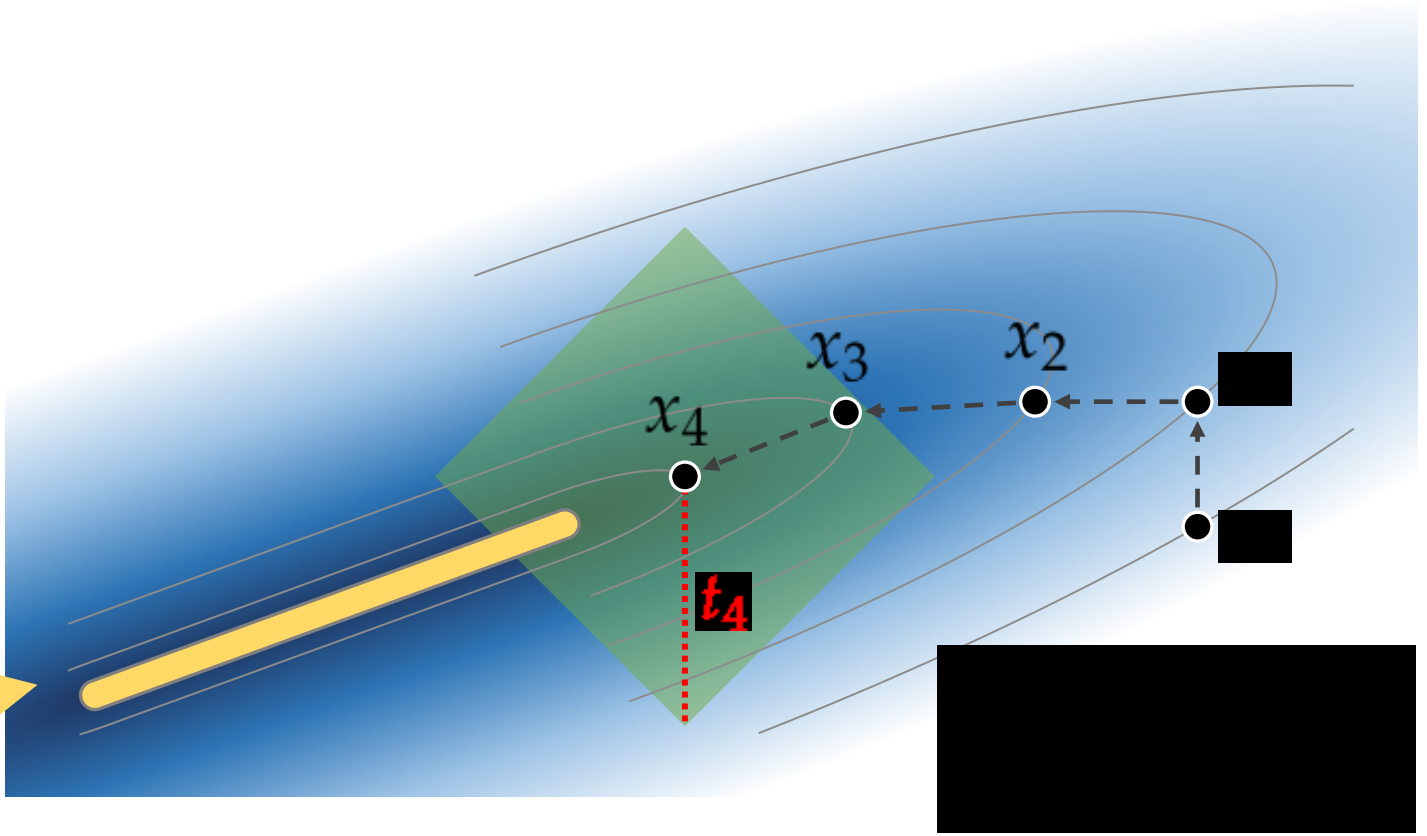
(

[redacted]

[redacted]

t_0	$d_1 = 6.04 \uparrow$		
$t_1 = 1.3$			
$t_2 = 1.6$			
$t_3 = 1.8$			

[redacted]



(ℓ_1 (&&

[redacted]

1 t_k

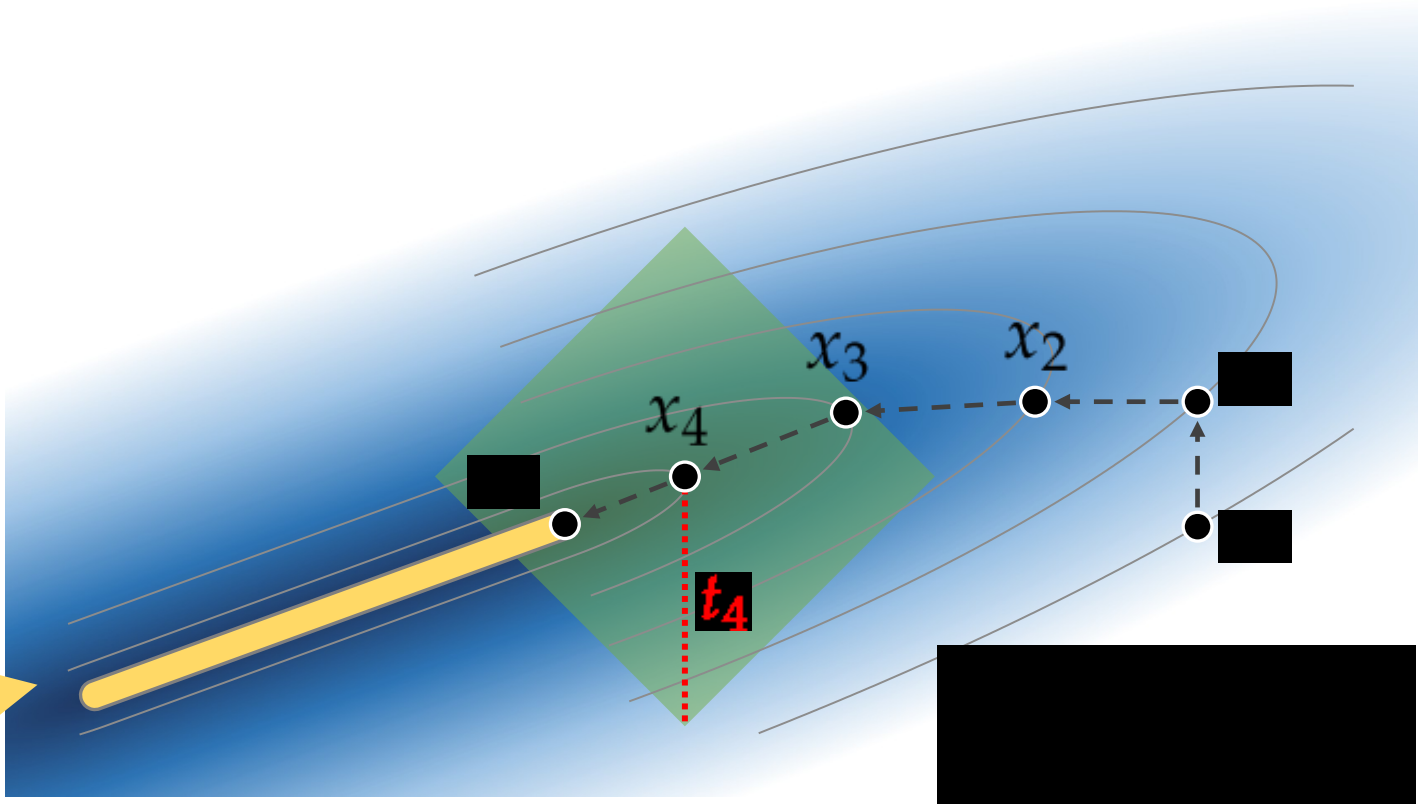
[redacted]

(

[redacted]

[redacted]			
t_0	$d_1 = 6.04$ ↑		
$t_1 = 1.3$			
$t_2 = 1.6$			
$t_3 = 1.8$			
t_4			

[redacted]



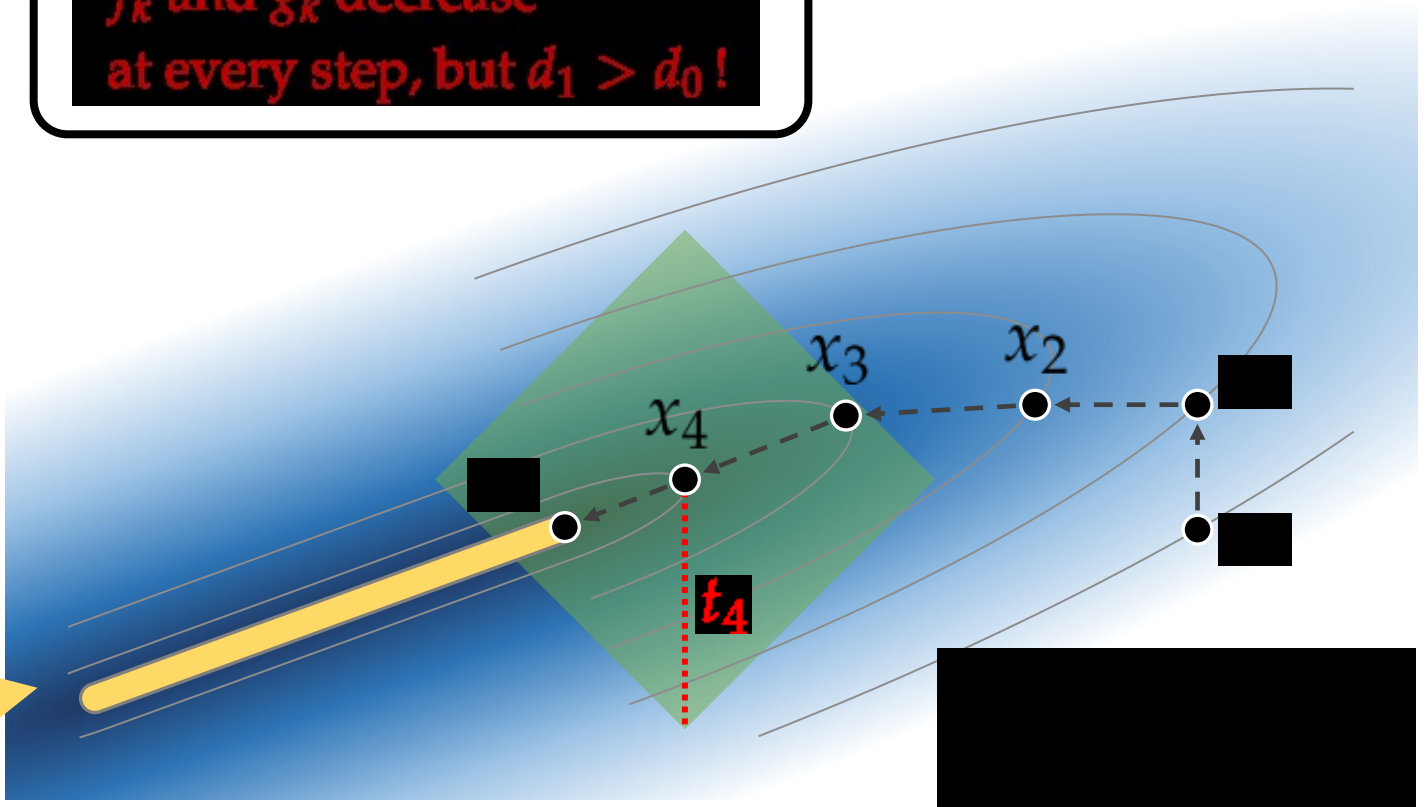
(ℓ_1 (&&

1 t_k

(

t_0	$d_1 = 6.04 \uparrow$		
$t_1 = 1.3$			
$t_2 = 1.6$			
$t_3 = 1.8$			
t_4			

f_k and g_k decrease at every step, but $d_1 > d_0$!



(ℓ_∞ (&&

∞ t_k

(ℓ_∞ (&&

∞ t_k

